

# Marginal Utility Shocks and the Precautionary Saving Puzzle

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## Abstract

This paper introduces a novel mechanism linking wealth accumulation to shocks in marginal utility rather than income alone. In contrast to the Bewley–Aiyagari–Huggett (BAH) framework, where savings eventually decline with wealth, I show that when households face idiosyncratic longevity shocks that alter the continuation value of utility, saving can remain strictly positive above some threshold. The model predicts that maintaining marginal utility becomes a luxury relative to consumption, inducing precautionary saving even among the wealthy. Using Health and Retirement Study (HRS) data on U.S. couples aged 55–65, I document that saving and saving rates rise with wealth, contrary to BAH predictions. Calibrating the model with health and income transitions—using imputed total medical expenditure to identify the health production function—I find that the mechanism accounts for approximately 31 percent of observed average savings in the top wealth decile.

**Keywords:** Precautionary Saving, Marginal Utility Shocks, Wealth Inequality, Incomplete Markets, Health Expenditure, Luxury Consumption, Heterogeneous Agents, Saving Function, Health and Retirement Study (HRS), Stationary Wealth Distribution

# 1 Introduction

Standard macroeconomic models with complete markets, such as those in Ljungqvist and Sargent (2012), fail to generate precautionary saving behavior or wealth inequality. In these frameworks, perfect consumption smoothing eliminates any incentive to save for uncertain future events, and the homogeneity in agents' starting conditions precludes the emergence of wealth heterogeneity.

To address this, a class of models introduced by Bewley (1977), Aiyagari (1994), and Huggett (1993)—commonly referred to as BAH models—incorporate market incompleteness and idiosyncratic income risk. These models succeed in generating precautionary savings and wealth dispersion due to the lack of full insurance. However, they still fall short in replicating the empirical fat-tailed wealth distribution observed in the United States (Saez and Zucman (2016)), Canada (Statistics Canada (2025)), and Europe (OECD (2024)). Even when calibrated with highly skewed income processes, BAH models predict thin tails because high-wealth individuals can still smooth consumption effectively, eliminating incentives to continue saving.

Several extensions to BAH models have been proposed to better match empirical data, including heterogeneity in time preferences (Cagetti (2003)), luxury bequests (De Nardi (2004)), return heterogeneity (Benhabib et al. (2011, 2015)), entrepreneurship (Cagetti and De Nardi (2006)), and health shocks (De Nardi et al. (2010)). Nevertheless, as De Nardi (2017) suggests, further progress requires endogenizing decision variables and exploring new mechanisms.

This paper proposes a novel mechanism wherein shocks to marginal continuation value, particularly in the form of longevity and health shocks, give rise to a distinct form of precautionary saving. Here, preserving marginal utility becomes a luxury relative to consumption, leading individuals to save not for consumption smoothing but to maintain utility over time—a departure from the canonical BAH logic. This results in a saving function that may remain strictly positive, even at high levels of wealth.

Moreover, this framework offers a compelling explanation for certain empirical regularities—most notably, the persistent increase in health spending, which is typically exogenous in standard models. By treating health maintenance as a luxury good tied to survival, we

can rationalize the rising share of health expenditure observed in macro data (Hall and Jones (2007); Ales et al. (2014); Ozkan (2014); French and Kelly (2016); Dickman et al. (2016, 2017)).

We apply this model to analyze savings behavior among elderly couples using data from the Health and Retirement Study (HRS), providing new evidence on how wealthy households save in the face of longevity risk, even near retirement. This challenges traditional explanations centered on return heterogeneity and highlights the importance of precautionary motives at the top of the wealth distribution.

Strikingly, our calibrated model accounts for approximately 31 percent of the observed savings among top-decile households. This degree of explanatory power—derived solely from precautionary saving in response to marginal utility shocks—is quantitatively significant. It suggests that risk to future utility (particularly due to health-related uncertainty) may be a primary driver of upper-tail saving, complementing alternative explanations based on rate-of-return heterogeneity or bequest motives.

By endogenizing survival-driven utility preservation as a form of luxury consumption, our framework sheds new light on both the shape of the saving function and the composition of wealth at the top. These findings help reconcile empirical patterns in savings and wealth inequality with theoretical models that incorporate realistic health and longevity risk.

## **A Review of Literature**

When markets are complete, no incentives exist for precautionary saving ((Ljungqvist and Sargent, 2012)). Perfect consumption smoothing is possible and since people born alike, these models cannot generate heterogeneity. Therefore wealth inequality is absent in these models.

Bewley (1977); Aiyagari (1994); Huggett (1993) (BAH) introduce market incompleteness, while individual's wealth depends on entire history of idiosyncratic shocks. In these models, lack of full risk-sharing creates precautionary saving motive against uninsurable idiosyncratic income risk, which gives rise to heterogeneity in wealth. BAH models collectively became workhorse models of inequality research in modern macro (aka “income fluctuations models” Schechtman (1976); Schechtman and Escudero (1977)).

But BAH models cannot account for inequality in the top. Wealth distribution is Pareto in US Saez and Zucman (2016), Canada Statistics Canada (2025), Europe OECD (2024). Even for extreme distributions of idiosyncratic risk, BAH models predict a thin-tailed wealth distribution. Underlying reason here is even without Arrow securities, it becomes possible to smooth consumption “almost perfectly” with “large enough” wealth. Therefore, BAH models predict a threshold after which individuals cease saving—saving becomes negative. In addition, life cycle versions of BAH Huggett (1996) end up with the same conclusion.

Extensions of BAH introduce different channels to match the observed wealth distribution De Nardi (2017). These include luxury bequests De Nardi (2004), heterogeneous time preferences Cagetti (2003), earning risk Castaneda et al. (2003), medical expenses De Nardi et al. (2010), heterogeneity in rates of return Benhabib et al. (2011, 2015), entrepreneurship Cagetti and De Nardi (2006). However, as denoted by De Nardi (2017), one direction of future research should be toward endogenizing decision variables<sup>1</sup>.

However, the current literature does not consider maintaining the welfare achieved through consumption as a luxury. In our mechanism, shock affects marginal continuation value, in a way that keeping marginal utility from dropping becomes “luxury” compared to consumption (*à la* (Jones et al., 2019)). Therefore, it is optimal to maintain marginal utility, instead of smoothing consumption. Hence, Saving can be bounded away from zero, for certain parameter values.

Finally, our framework can be applied to explain different empirical observations. One example that our framework can explain is the rising share of health spending. In usual wealth distribution framework spending on health comes exogenously and therefore is not affected by the increase in available resources. However as documented by Hall and Jones (2007); Ales et al. (2014); Ozkan (2014); French and Kelly (2016); Dickman et al. (2016, 2017), share of health spending keeps rising in aggregate level, which implies the luxury nature of survival. In what follows, we use our model to explain saving behavior of couples observed in HRS.

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<sup>1</sup>See for example Gomez and Gouin-Bonenfant (2024) for a case of endogenous investment.

## 2 Motivation

The recent literature on saving keeps documenting the fact that saving becomes either flat Bach et al. (2016) or increasing Fagereng et al. (2019) at the top using detailed administrative data in different economies. Utilizing the health and retirement study (HRS), we provide another evidence that supports the notion of increasing saving policy by households in the top decile of wealth. The Health and Retirement Study (HRS) is sponsored by the National Institute on Aging (grant number NIAU01AG009740) and conducted by the University of Michigan. Surveys are collected every two years at the household level at the end of each survey year, beginning in 1992. Our research shows that while both saving and saving rate increases for couples in the top decile, the correlation with income drops for the pre-retirement elderly.

Figure 1 reports average savings of couple households in different wealth brackets among the US elderly from 1996 to 1998 in thousands of dollars Where saving is defined by the first difference of wealth. The first seven wealth groups report savings between \$27,000 and \$45,000, with these figures rising to \$72,000 and \$78,000 for the eighth and ninth wealth groups, respectively. A substantial increase in savings is observed for the wealthiest couples, with households in the top group accumulating approximately \$1,092,000 in net worth on average, which is not unexpected since the distribution of wealth is Pareto.<sup>2</sup>

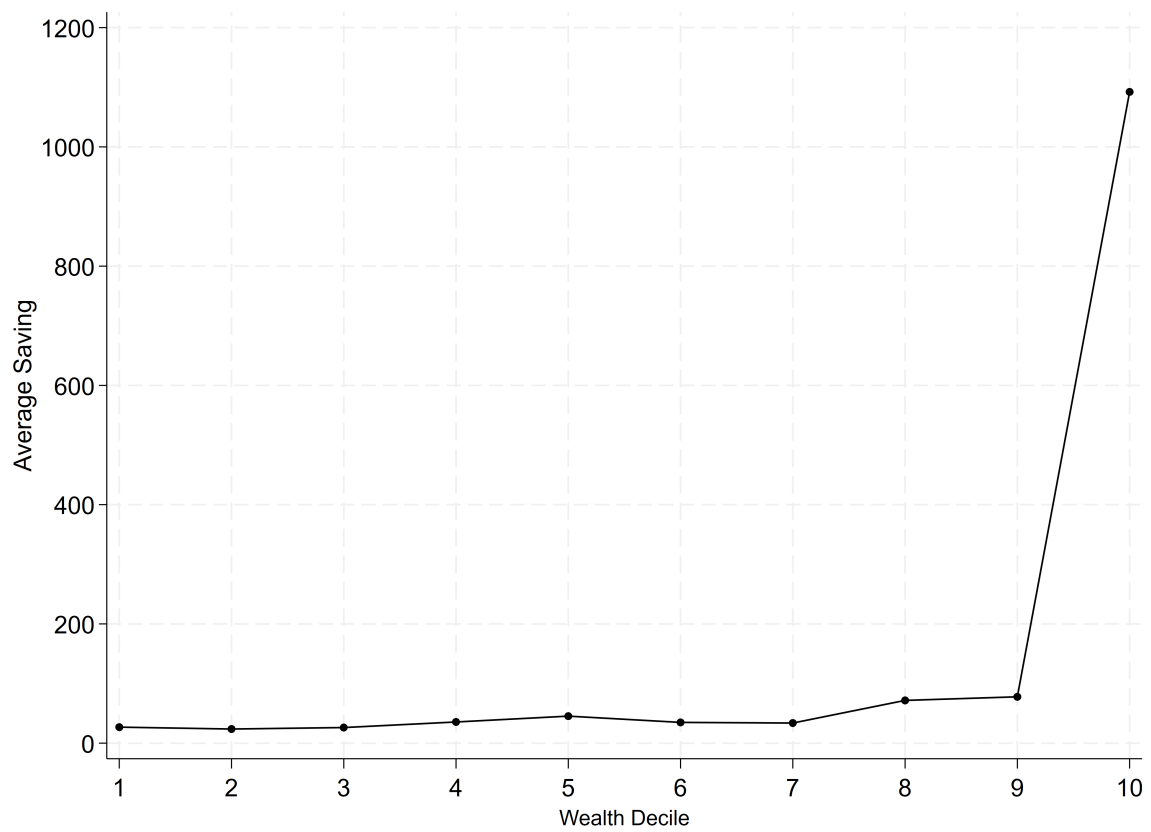
Among households in the top decile of the wealth distribution, income and saving rates are positively correlated, contrary to the predictions of standard Bewley–Aiyagari–Huggett models. Moreover, unlike the results in Fagereng et al. (2019); Bach et al. (2017), the correlation between saving rates and wealth returns declines for these top-decile couples. Since these households are near retirement, this non-monotonic pattern suggests that inequality at the top may stem from precautionary saving behavior among the wealthy rather than from idiosyncratic return heterogeneity, as in Benhabib et al. (2015, 2017).<sup>3</sup>

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<sup>2</sup>See Appendix B.3 for cohort selection, Appendix B.4.1 for the wealth distribution, and Appendix B.4.2 for saving patterns of singles in addition to couples.

<sup>3</sup>See Appendix B.4.3 for saving rates, and Appendix B.4.5 for income and saving rate correlation.

**FIGURE 1. Average Saving**



*This figure represents average savings of couple households in different wealth brackets among the US elderly from 1996 to 1998. Average savings are in thousands of dollars.*

### 3 Saving at the Tail

To understand the reason our framework can give rise to a saving function that is increasing in wealth, after some threshold—in contrast to the Bewley (1977); Aiyagari (1994); Huggett (1993) (BAH)’s framework where saving becomes negative after some threshold—we consider two stylized environments with uninsurable idiosyncratic shocks in this section: An income fluctuations problem, *a lá* BAH, where individuals have to self-insure against fluctuations in their income, and a modified framework where income shocks are replaced by longevity shocks—or, more accurately, shocks to continuation marginal utility.

Consider the problem of an individual in a standard income fluctuations framework, where income can take one of two values,  $\{\underline{y}, \bar{y}\}$ , where  $\underline{y} < \bar{y}$ . If we refer to these two as the “low” and “high states,” and denoting the value and policy functions in each state by means of lower and upper bars, respectively, we can write the individual’s problem in high state as

$$\begin{aligned} \bar{V}(b) = \max_{c, b'} \quad & \{u(c) + \rho [\pi \bar{V}(b') + (1 - \pi) \underline{V}(b')]\} \\ \text{s.t.} \quad & b' = (1 + r)b + \bar{y} - c, \\ & b' \geq \underline{b}. \end{aligned} \tag{BAH}$$

In this problem,  $\pi$  is the conditional probability of staying in the high state in the next period, conditioned on being in the high state in the current period. The problem in the low state is rather similar to (BAH), and we omit it here.

In our framework, we replace income shocks by longevity shocks. To this end, we denote an individual’s underlying health by  $h$ , and assume it can take two values,  $\{\underline{h}, \bar{h}\}$ , where  $\underline{h} < \bar{h}$ . With some abuse of terminology, we will refer to these two as the “low” and “high states,” and keep denoting them via lower and upper bars as in problem (BAH).<sup>4</sup>

To make sure the proceeding arguments deal explicitly and entirely with these two different mechanisms, in this section, we assume that the low state is an absorbing state in this latter

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<sup>4</sup>This choice of notation serves a simplifying purpose, as it becomes apparent when writing the first order conditions. It should be clear which problem a function refers to from the context. We make the reference explicit if not.

framework: Once an individual enters the low state, he remains there forever. The probability of an individual transitioning to this state from the high state is denoted by  $1 - \pi$ , as in problem (BAH). In addition, we assume that an individual does not face any chance of mortality, whereas this probability in the low state is determined by the *health production function*,  $\chi(m)$ , where  $m$  captures what the individual spends on his health.

As such, the individual's problem in the high state is quite similar to that in (BAH) in this alternative setting:

$$\begin{aligned} \bar{V}(b) = \max_{c, b'} \quad & \{u(c) + \rho [\pi \bar{V}(b') + (1 - \pi) \underline{V}(b')]\} \\ \text{s.t.} \quad & b' = (1 + r)b + y - c, \\ & b' \geq \underline{b}, \end{aligned} \tag{BL-H}$$

where  $y$  denotes the individuals' income, which is independent of their health state (so that we can focus on marginal utility shocks, as states earlier). In the low state, this problem becomes:

$$\begin{aligned} \underline{V}(b) = \max_{c, m, b'} \quad & \{u(c) + \rho \cdot \chi(m) \cdot \underline{V}(b')\} \\ \text{s.t.} \quad & b' = (1 + r)b + y - c - m, \\ & b' \geq \underline{b}. \end{aligned} \tag{BL-L}$$

Note that, in this problem, we have implicitly normalized the value upon death to zero.

In all that follows, we are going to assume that the Bernoulli utility takes the CRRA functional form with a constant term representing the value of being alive:

$$u(c) = \nu + \frac{c^{1-\sigma}}{(1-\sigma)},$$

for some  $\sigma > 0$  and  $\nu > 0$  such that, for some minimal sustainable level of consumption,  $\underline{c}$ , we have  $u(\underline{c}) > 0$ .<sup>5</sup> Moreover, we assume the following form for the health production

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<sup>5</sup>Otherwise, (BL-L) turns into a trivial problem where individuals prefer death to being alive. The CRRA utility form is indispensable to Huggett (1993)'s proof of negative saving above some level.

function for the sake of tractability:

$$\chi(m) = 1 - \left( \frac{1}{\alpha \cdot m + \underline{h}} \right)^\beta, \quad (1)$$

for some  $\alpha > 0$  and  $\beta \leq 1$ .

Consider the Euler equation for an *interior solution* to problems (BAH) and (BL-H):

$$\bar{V}_b(b) = \rho \cdot (1 + r) \cdot [\pi \bar{V}_b(b') + (1 - \pi) \underline{V}_b(b')], \quad (\text{HEE})$$

where subscripts show first order derivatives, and we have used the envelope condition to replace for marginal utility of consumption:

$$\bar{V}_b(b) = (1 + r) \cdot u_c(c). \quad (\text{HEC})$$

Under the assumption that the value function is concave in wealth in both states, the Euler equation determines the direction of saving at any current level of wealth. To see this more clearly, let us momentarily assume that  $\pi = 1$ . When so, when  $\rho(1 + r) = 1$ , we must have  $b = b'$  and saving will be zero:  $c = c' = rb + \bar{y}$  (or  $c = c' = rb + y$ , when income is constant). In other words, if time discount rate and the rate of return cancel out, individual would “smooth consumption” perfectly. (When  $\rho(1 + r) < 1$ , individual front loads consumption: In this case, for (HEE) to hold,  $\bar{V}_b(b)$  must be greater than  $\bar{V}_b(b')$ . Assuming that the value function is concave, we must have  $b' < b$  for this to be the case. The inverse of this argument holds when  $\rho(1 + r) > 1$ .)

In the presence of uninsurable risk, saving might not be negative at lower levels of wealth even when  $\rho(1 + r) < 1$ , as long as the arrival of the shock (whether it is to income or underlying health) significantly increases the marginal effect of assets on future value. We will refer to this—i.e.  $\underline{V}_b(b')$ —as the *marginal continuation value*. To see this, let us write (HEE) as

$$\pi \left[ \frac{\bar{V}_b(b')}{\bar{V}_b(b)} \right] + (1 - \pi) \left[ \frac{\underline{V}_b(b')}{\bar{V}_b(b)} \right] = \frac{1}{\rho \cdot (1 + r)}. \quad (2)$$

When value function is concave, the first term in brackets in this equation will be less than one when  $b' > b$ . Therefore, the only possibility for this equality to hold when  $b' > b$  and

$\rho(1+r) < 1$  is for  $\underline{V}_b(b')$  to be greater than  $\bar{V}_b(b)$  by a “large enough” margin, and this is the logic behind the terminology “precautionary saving”: If the value of having one more unit of wealth is high enough *when an unfavourable shock is realized*, individual tends to save.

One can summarize this argument as follows: Under the concavity assumption for the value function, saving is positive at any current level of wealth  $b$  if, and only if,

$$\pi \left[ \frac{\bar{V}_b(b)}{\bar{V}_b(b)} \right] + (1 - \pi) \left[ \frac{\underline{V}_b(b)}{\bar{V}_b(b)} \right] > \frac{1}{\rho \cdot (1 + r)}.^6$$

This inequality can be rearranged as

$$(1 - \pi) \left[ \frac{\underline{V}_b(b) - \bar{V}_b(b)}{\bar{V}_b(b)} \right] > \frac{1}{\rho \cdot (1 + r)} - 1. \quad (3)$$

The right-hand side of this equality is a positive constant when  $\rho(1+r) < 1$ . As such, the shape of optimal saving as a function of current wealth,  $b$ , is determined by the marginal value of wealth in the low state *relative to that in the high state* (and not by the difference between the two).

This is rather intuitive: Individual saves as a “precaution” against the possibility of low shocks, and as long as the marginal benefits of one additional unit of saving is high enough should the low shock arrives. Saving is a precaution against fluctuations in marginal continuation value when value function is concave. On the other hand, when the ratio of marginal continuation value in the low state to that in the high state tends to one, this precautionary saving motive vanishes: When there are not much fluctuations in the marginal continuation value in the first place, there are no incentives to insure against such fluctuations either.<sup>7</sup>

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<sup>6</sup>Similarly, saving will be zero or negative, respectively, when wealth is equal to  $b$  if, and only if, this inequality holds as strict equality or the left-hand side is strictly less than the constant term on the right-hand side.

<sup>7</sup>This is not to say that the only incentive for saving is precautionary motive. The ratio of time discount rate to the risk-free rate is still a strong determinant of saving behaviour. Moreover, the precautionary saving motive is determined by the probability of the realization of unfavourable shocks.

One can use condition (HEC) to write (3) as

$$(1 - \pi) \left[ \frac{u_c(\underline{c}(b)) - u_c(\bar{c}(b))}{u_c(\bar{c}(b))} \right] > \frac{1}{\rho \cdot (1 + r)} - 1, \quad (4)$$

where  $\underline{c}(\cdot)$  and  $\bar{c}(\cdot)$  are the policy functions in low and high states, respectively. If we replace the utility function with its CRRA form, we can write this inequality as:

$$\left[ \frac{\bar{c}(b)}{\underline{c}(b)} \right]^\sigma > \left[ \frac{\rho^{-1}(1 + r)^{-1} - \pi}{1 - \pi} \right]. \quad (5)$$

Note that the first term of the numerator on the right-hand side of this inequality is greater than one. As such, the right-hand side is a constant, greater than one.

To see the intuition behind this inequality, we can use a linear expansion around  $\bar{c}(b)$  to write the numerator in (4) in terms of consumption in two states:

$$(1 - \pi) \left[ \frac{u_{cc}(\bar{c}(b)) [\underline{c}(b) - \bar{c}(b)]}{u_c(\bar{c}(b))} \right] > \frac{1}{\rho \cdot (1 + r)} - 1.$$

Using the CRRA utility form, the ratio on the left-hand side of this inequality can be written as:

$$1 - \frac{\underline{c}(b)}{\bar{c}(b)} > \frac{1 - \rho \cdot (1 + r)}{\sigma \cdot (1 - \pi) \cdot \rho \cdot (1 + r)}.^8 \quad (6)$$

As such, the shape of saving function depends critically on  $\sigma$ ,  $\pi$ , and  $\rho(1 + r)$ , as well as the ratio  $\underline{c}(b)/\bar{c}(b)$  given these parameters. As this inequality suggests, if  $1 - \pi$  or  $\sigma$  are too small, it is well possible for saving to remain negative at all levels of wealth. This is rather intuitive: When the possibility of a negative shock is “too small,” or when individuals are “too risk neutral,” precautionary saving motive would also be too small. In such cases, if the incentive to front-load consumption is strong enough—when  $\rho(1 + r)$  is far from unity—individual might not have any incentives to save against unlikely adverse shocks in the future.

On the other hand, when  $1 - \pi$  and  $\sigma$  are large enough, it is possible for the precautionary

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<sup>8</sup>Note that, since  $u_c(\cdot)$  itself is a convex function, this inequality is the sufficient condition for the saving rate to be positive and not a necessary condition, at least unless consumption is not “too large.”

saving motive to dominate the motive to front-load consumption, leading to positive saving. When this is the case, we'd expect the ratio of consumption in two states to remain far from one. The same forces that determine how this ratio changes as wealth increases (in the optimal solution) also determine the shape of optimal saving as a function of wealth.

The aforementioned arguments are summarize in lemma 3.1, under the assumption that the solutions to the functional equations in (BL-H) and (BL-L) are concave. This assumption, however, cannot be proved analytically: The product of two concave functions on the right-hand side of functional equation (BL-L) is not necessarily concave. As a result, the usual inheritance arguments in dynamic programming do not carry over to this case. As such, throughout this section, we maintain the following assumption:

**ASSUMPTION 3.1** *The solution to the functional equation (BL-L),  $\underline{V}(\cdot)$ , is concave.*

Note that, if  $\underline{V}(\cdot)$  is concave,  $\bar{V}(\cdot)$  will necessarily be concave. Moreover, we know from the envelope condition that assumption 3.1 is equivalent to the policy function,  $\underline{c}(\cdot)$ , being increasing. In our numerical experiments, we have not managed to come up with a set of parameter values that violate this latter condition.

**LEMMA 3.1** *Under assumption 3.1, optimal saving in (BAH) or (BL-H) is positive when wealth is  $b$  if, and only if, the ratio of optimal consumption in high state to that in low state satisfies (5).*

The preceding argument suggests that, to determine the shape of saving as a function of current wealth in either problem, one can focus on the ratio  $\bar{c}(b) / \underline{c}(b)$  and how it changes with wealth. One can show that, in problem (BAH), this ratio converges to one as  $b \rightarrow \infty$ , and, as such, there exists some  $\bar{b}$  above which saving inevitably becomes negative.<sup>9</sup>

In our environment with health shocks, on the other hand, the same mechanics but in the opposite direction indicate the exact inverse regarding the saving behaviour: that optimal saving may have to be positive above some threshold  $\bar{b}$  (while it can be negative below  $\bar{b}$ ).

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<sup>9</sup>This claim, stated by Huggett (1993) in Theorem 2, enables him to show that an individual's wealth does not exceed  $\bar{b}$ . As such, the state-space is bounded and a stationary distribution of wealth can emerge as a result. We postpone the proof of this claim for problem (BAH)—which is different from Huggett (1993)'s proof—to appendix A.

The reason that the ratio  $\bar{c}(b) / \underline{c}(b)$  may be bounded away from one in problem (BL-H) is due to the role of health spending in (BL-L).

The intuition is as follows: If health spending is a luxury good, its share in total expenditures must increase with wealth compared to that of consumption. In other words, in the presence of this luxury good, the marginal value of saving remains high in the low state compared to that in the high state, even as individual becomes wealthier. As such, the precautionary saving motive does not vanish with wealth, as is the case in the income fluctuation problem in (BAH).

To formally derive the conditions under which inequality (5) holds, at least above some wealth threshold  $\bar{b}$ , we start by characterizing the optimal medical spending in the low-state, in (BL-L). When interior, the optimal health spending is given by the following first order condition:

$$\rho \cdot \chi_m(m) \cdot V(b') = u_c(c).$$

Replacing the utility and health production functions by their functional forms, we can rearrange this equation to find optimal health spending as a function of optimal consumption in the low state. This is given by

$$m(b) = \left[ \frac{\rho \cdot \beta \cdot V(b'(b))}{\alpha^\beta} \right]^{\left(\frac{1}{1+\beta}\right)} \cdot [\underline{c}(b)]^{\left(\frac{\sigma}{1+\beta}\right)} - \left(\frac{h}{\alpha}\right), \quad (7)$$

when the right-hand side of (7) is positive, and zero otherwise. In this equation,  $\underline{m}(\cdot)$ ,  $\underline{c}(\cdot)$  and  $\underline{b}'(\cdot)$  are the policy functions for medical spending, consumption and future wealth in the low state, respectively.

We are now in a position to prove the following theorem, the central result of this paper:

**THEOREM 3.2** *Under assumption 3.1, saving rate in problem (BL-H) cannot be negative above some threshold  $\bar{b}$ , when  $\sigma > 1 + \beta$ .*

*Proof.* We accept, without proof, that the value function in (BL-L) is non-decreasing in cur-

rent wealth. As such, we can use (7) to conclude:

$$\begin{aligned}
\underline{m}(b) &= \left[ \frac{\rho \cdot \beta \cdot V(\underline{b}(b))}{\alpha^\beta} \right]^{\left(\frac{1}{1+\beta}\right)} \cdot [\underline{c}(b)]^{\left(\frac{\sigma}{1+\beta}\right)} - \left(\frac{h}{\alpha}\right) \\
&\geq \left[ \frac{\rho \cdot \beta \cdot V(\underline{b})}{\alpha^\beta} \right]^{\left(\frac{1}{1+\beta}\right)} \cdot [\underline{c}(b)]^{\left(\frac{\sigma}{1+\beta}\right)} - \left(\frac{h}{\alpha}\right) \\
&= A \cdot [\underline{c}(b)]^{\left(\frac{\sigma}{1+\beta}\right)} - \left(\frac{h}{\alpha}\right), \quad (8)
\end{aligned}$$

where

$$A := \left[ \frac{\rho \cdot \beta \cdot V(\underline{b})}{\alpha^\beta} \right]^{\left(\frac{1}{1+\beta}\right)} > 0.$$

The claim that  $A > 0$  follows from the assumption that, even at  $b = \underline{b}$ , individual can maintain a sustenance level of consumption, delivering a utility that is strictly greater than zero—*i.e.* the utility upon death. When  $\sigma > 1$ , the value of being alive,  $\nu > 0$ , guarantees that this is the case. Since such a stream of consumption is sustainable as long as the individual lives, value function must remain strictly positive for all  $b$ .

Let us assume, for the sake of contradiction, that saving is non-positive above some arbitrary threshold  $\tilde{b}$ . Then, we must have

$$\bar{c}(b) \geq r \cdot b + y, \quad \forall b \geq \tilde{b}.$$

If we let

$$\gamma := \liminf_{b \rightarrow \infty} \frac{\bar{c}(b)}{\underline{c}(b)},$$

condition (5) implies that

$$\gamma \leq \left[ \frac{\rho^{-1}(1+r)^{-1} - \pi}{1 - \pi} \right]^{\frac{1}{\sigma}}, \quad (9)$$

above some threshold of wealth. Therefore, we can assume without any loss of generality that, for all  $b \geq \tilde{b}$ ,

$$\underline{c}(b) \geq \gamma \cdot \bar{c}(b) = \gamma(r \cdot b + y), \quad (10)$$

and—from (8),

$$\underline{m}(b) \geq A[\gamma \cdot \bar{c}(b)]^{\left(\frac{\sigma}{1+\beta}\right)} - \left(\frac{h}{\alpha}\right) \geq A[\gamma(r \cdot b + y)]^{\left(\frac{\sigma}{1+\beta}\right)} - \left(\frac{h}{\alpha}\right). \quad (11)$$

If we replace these in individual's budget constraint, we get:

$$\begin{aligned} \underline{b}'(b) &= (1+r)b + y - \underline{c}(b) - \underline{m}(b) \\ &\leq (1+r)b + y + \left(\frac{h}{\alpha}\right) - \gamma(r \cdot b + y) - A[\gamma(r \cdot b + y)]^{\left(\frac{\sigma}{1+\beta}\right)} \\ &= -A[\gamma(r \cdot b + y)]^{\left(\frac{\sigma}{1+\beta}\right)} + (1+r-\gamma \cdot r)b + (1-\gamma)y + \left(\frac{h}{\alpha}\right). \end{aligned} \quad (12)$$

When  $\sigma > 1 + \beta$ , the polynomial on the right-hand side of this inequality is concave and, as such, decreases without bound with  $b$ . As a result, the borrowing constraint must bind for all  $b \geq \hat{b}$ , for some large enough  $\hat{b}$ . We let  $\bar{b}$  denote the maximum of this threshold and  $\tilde{b}$ :

$$\bar{b} := \max \left\{ \tilde{b}, \hat{b} \right\}.$$

The Kuhn-Tucker conditions for problem (BL-H) when the borrowing constraint binds imply that

$$u_c(c) = \rho \cdot \chi(m) \cdot \underline{V}_b(b') + \mu, \quad (13)$$

where  $\mu \geq 0$  is the Lagrange multiplier on the borrowing constraint, which is strictly positive when this constraint binds. Thus, for all  $b \geq \bar{b}$ , we should have:

$$u_c(c) = \chi(m) \cdot \rho \cdot \underline{V}_b(\underline{b}) + \mu. \quad (14)$$

As noted earlier, the term  $\rho \cdot \underline{V}_b(\underline{b})$  in this equality is a positive constant, independent of  $b$ . Moreover, from (10) and (11), we know that  $\underline{c}(b)$  and  $\underline{m}(b)$  both increase without bound with  $b$ . Therefore,  $u_c(\underline{c}(b)) \searrow 0$  and  $\chi(\underline{m}(b)) \nearrow 1$ , as  $b \rightarrow \infty$ . Therefore, as long as  $\mu$  is to remain positive, equation (14) is bound to get violated for some  $b \geq \bar{b}$ . This contradicts the optimality of  $\underline{c}(b)$  and  $\underline{m}(b)$ , and completes our proof.  $\square$

In the proof of theorem 3.2, what causes a contradiction is not whether or not  $\bar{c}(b) / \underline{c}(b)$

converges to a constant term *per se*. In fact, this ratio can theoretically converge to a constant even when saving remains positive for all  $b$  above some threshold  $\bar{b}$ , as long as this ratio is above the threshold prescribed by (5). What leads to the contradiction is the assumption that saving becomes negative for large enough  $b$ . As such,  $\bar{c}(b)$  must increase with  $rb + y$  to keep the saving negative. Moreover, negative saving implies that  $\bar{c}(b)$  and  $\underline{c}(b)$  cannot diverge. It is the combination of these two that lead to a contradiction. For saving to remain positive, then, we need the ratio  $\bar{c}(b) / \underline{c}(b)$  to remain above

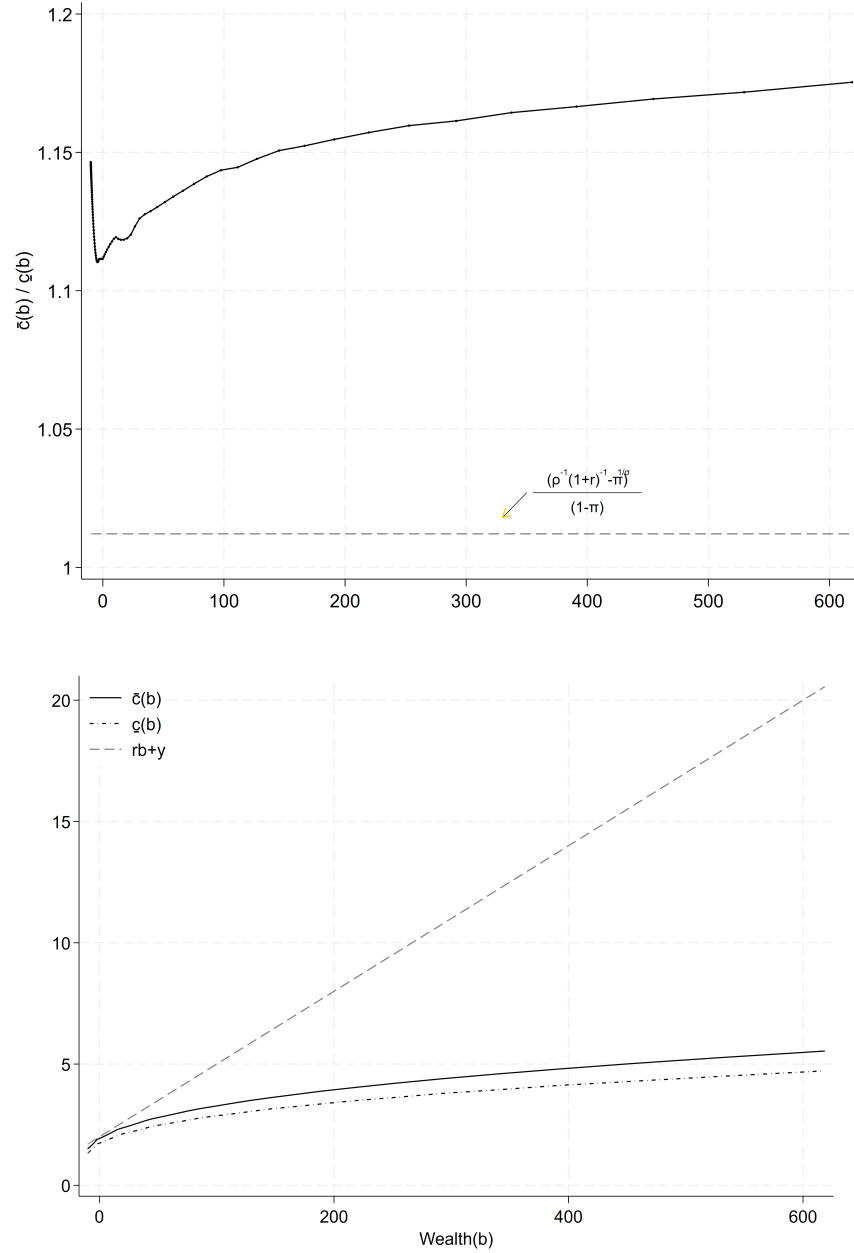
$$\left[ \frac{\rho^{-1}(1+r)^{-1} - \pi}{1 - \pi} \right]^{\frac{1}{\sigma}}.$$

As figure 2 demonstrates how, for our sample choice of parameter values, consumption in the high state diverges from total income, leading to excessive saving. Panel (b) in this graph suggests that the ratio  $\bar{c}(b) / \underline{c}(b)$  remains consistently above the right-hand side of (5). (In the numerical solutions to the problem in (BL-H) and (BL-L), this ratio seems to be exploding, though at a slow rate, as can be seen in figure 2 too.)

The condition  $\sigma > 1 + \beta$  in theorem 3.2 has an intuitive and straightforward interpretation: As long as the *curvature* of the marginal utility (and not the marginal utility itself) is greater than the *curvature* of the marginal product of health spending (in the health production function), then health is a luxury good. It is important to note that it is not the result that health spending is a luxury that causes precautionary saving to increase without bound, *per se*; But it is the fact that such spending is necessitated by an adverse health shock.

It is also worth mentioning that  $\sigma > 1 + \beta$  is not a necessary condition for health spending to increase faster than consumption with wealth, but a sufficient one: Any assumption implied by equation (7) that causes health spending to increase faster than consumption with  $b$  suffices for the proof of theorem 3.2 to go through. This can be a high enough value of being alive ( $\nu$ ) or a high enough share of health spending ( $\alpha$ ) in the health production function, as confirmed by our numerical results.

**FIGURE 2.** Consumption in Low and High States as Function of Wealth



**Note:** For  $\bar{c}(b) / \underline{c}(b)$ , centred moving averages are drawn to smooth out high-frequency fluctuations at lower wealth levels due to numerical error.

## 4 Stationary Distribution

In this section, we show how the combined effect of saving at the tail and random separation (i.e., mortality) can lead to a stationary Pareto distribution of wealth in a partial equilibrium setting.

To this end, consider an economy in which, in each period  $t$ , a new cohort of individuals—of measure  $\psi_0$ —enters the economy with initial wealth distribution  $\varphi_0(\cdot)$  (where  $\varphi_0(\cdot)$  is a probability measure, defined on  $(S, S)$ , in which  $S$  is a Borel subset of  $\mathbb{R}$ ), in high health.

An individual's problem is the same as before, with healthy individuals facing a probability  $1 - \pi$  of transition to the low health state, and individuals in low health with medical spending  $m$  facing probability  $1 - \chi(m)$  of mortality.

The distribution of wealth in this economy can be written as the sum of the conditional distribution of wealth for individuals in high and low health states, weighted by the relative measure of each. If one of these distributions follows a power law—e.g., is Pareto—then the sum of the two will also follow a power law, even if the other one does not. As noted by Gabaix (2009), *Power Laws in Economics and Finance*:

“Power laws also have excellent aggregation properties. [...] The general rule is that, when combining two PL variables, the fattest (i.e., the one with the smallest exponent) PL dominates. [...] Hence, multiplying by normal variables, adding nonfat tail noise, or summing over independent and identically distributed (i.i.d.) variables preserves the exponent.”

As such, in what follows, we focus on the tail behaviour of the conditional distribution of wealth for high-health individuals, and show that this distribution follows a power law with a power exponent equal to one—i.e., it follows Zipf's law at the tail.

**LEMMA 4.1** *Any discrete-time, heterogeneous-agent economy with random separation has a stationary state-distribution.*

*Proof.* Consider an economy in which individuals are heterogeneous in some dimension—say  $b$ , where  $b$  can be any vector of states. Suppose an initial measure  $\psi_0$  of individuals enter

the economy in every period, with initial distribution  $\varphi_0(\cdot)$ . Of these individuals, fraction  $1 - \pi$  separate in each subsequent period.

Assuming  $P$  is the economy's transition function and  $T^*$  the adjoint operator associated with it, the distribution of individuals who enter in  $t$ , in  $t + n$  would be

$$T^{*n}\varphi_0(\cdot) = \varphi_n(\cdot).$$

Then, for any  $s \in S$ , we must have:

$$\left(\frac{\psi_0}{1 - \pi}\right) \cdot \varphi(s) = \psi_0[\varphi_0(s) + \pi \cdot \varphi_1(s) + \pi^2 \cdot \varphi_2(s) + \pi^3 \cdot \varphi_3(s) + \dots].$$

Since  $\varphi_n(\cdot) < 1$  for all  $n$  (and  $\pi < 1$ ), this infinite sum is bounded and converges for all  $s$ . □

**THEOREM 4.2** *When  $b' \geq (1 + \gamma)b$ , for some  $\gamma > 0$ , the stationary distribution of wealth follows Zipf's law at the upper tail.*

*Proof.* Let saving be positive for some  $x > \underline{b}$ , and assume  $b' = (1 + \alpha)b$  for all  $b \geq x$ . Because we are only interested in the tail behaviour of the wealth distribution, we assume—without any loss of generality—that  $S = [x, +\infty)$ .

Since a stationary distribution exists, the above-mentioned assumption on saving implies:

$$\Pr(b \geq x) = \pi \cdot \Pr\left(b \geq \frac{x}{1 + \alpha}\right).$$

Also, for  $x$ , we must have  $\Pr(x \geq x) = 1$ . If  $n$  is the smallest integer for which  $x \leq (1 + \alpha)^n \cdot x$ , then

$$\Pr(b \geq x) \geq \pi^n \cdot \Pr(x \geq x) = \pi^n.$$

Writing  $n$  as

$$n \geq \frac{\ln(x/x)}{\ln(1 + \alpha)},$$

we have

$$\ln \Pr(b \geq x) \geq n \ln(1 - \pi) \geq \frac{\ln(1 - \pi)}{\ln(1 + \alpha)} \cdot \ln\left(\frac{x}{x}\right).$$

Define

$$\gamma = -\frac{\ln(1 - \pi)}{\ln(1 + \alpha)} > 0.$$

Thus,

$$\ln \Pr(b \geq x) \geq \gamma \cdot \ln\left(\frac{x}{x}\right),$$

or equivalently,

$$\Pr(b \geq x) \geq \left(\frac{e^\gamma \cdot x}{x}\right).$$

This shows that the distribution of wealth in this economy follows Zipf's law in its tail, regardless of what the conditional distribution of wealth for low-health individuals looks like.  $\square$

## 5 Empirical Model & Calibration

We extend the framework by introducing income shocks alongside longevity shocks as an additional source of heterogeneity. The transition matrix is denoted  $\Gamma_y$ . Intuitively, at the initial date a mass  $\psi_0$  of agents possess high health. In period 1, a fraction transitions to the low-health state and remain there. Within this group, there are low-income agents with poor health, and high-income agents with poor health. The analogous stationary distribution argument from the previous section applies here.

Survival is independent of income but depends on health status:

$$\chi(m, h) = 1 - \left(\frac{1}{\alpha m + h}\right)^\beta, \quad \forall h \in \mathcal{H}.$$

Unlike the baseline case, a fraction of high-health agents may exit with probability  $\chi(m, \bar{h})$ . The entry condition ensures that the mass of healthy individuals remains at  $\psi_0/(1 - \pi)$ , implying survival is effectively certain in the high-health state even with minimal medical spending; to ensure this, we set  $\bar{h} = 9999$ .

We next introduce public health insurance. Once a household moves to the low–health state, it receives a transfer  $\underline{m}$ , calibrated to \$647 thousand from the average long-term Medicaid payment. The household’s problem is

$$\begin{aligned}
V(b, y, h) = \max_{c, m, b'} & \left\{ u(c) + \rho \chi(\hat{m}, h) \mathbb{E}_{y', h'} [V(b', y', h')] \right\} \\
\text{s.t. } & b' = (1 + r)b + y - c - m, \\
& \hat{m} = \begin{cases} m + \underline{m}, & h = \underline{h}, \\ m, & h = \bar{h}, \end{cases} \\
& b' \geq -\underline{b}.
\end{aligned} \tag{FUL}$$

The borrowing limit is set to  $\underline{b} = 646.7$  thousand dollars, equal to the Medicaid transfer. This choice reflects the assumption that, in the worst case, a household entering the low–health state can borrow against the present value of future public insurance transfers. Since Medicaid provides a guaranteed floor of medical spending in the low–health state, the maximum amount a household can credibly pledge to repay is bounded by this transfer.

## Estimating the Health Production Function

The parameters  $\alpha$  and  $\beta$  in the health production function govern how medical spending translates into survival probability. These are estimated by matching model-implied survival to self-reported survival expectations and medical spending in the HRS via nonlinear least squares:

$$\min_{\alpha, \beta} \sum_i \left[ \chi_i - \left( 1 - \frac{1}{(\underline{h} + \alpha \ln m_i)^\beta} \right) \right]^2. \tag{EST}$$

The estimation uses log-transformed medical spending rather than levels. The raw level of total medical expenditure spans several orders of magnitude across households, from a few hundred dollars to several hundred thousand. Estimating in levels would allow a small number of extreme observations to dominate the nonlinear least squares objective. The log transformation compresses this scale, stabilizes the numerical optimization, and reduces the influence of outliers on the estimated parameters. Since both the survival function and the

log transformation are monotone, this reparametrization does not alter the qualitative properties of the health production function—in particular, the conditions under which health spending acts as a luxury good (Theorem 3.2) are preserved.

A key feature of this estimation is the use of imputed total medical expenditure (see Appendix C), rather than out-of-pocket spending or medical spending shares. The health production function relates survival to the *total* resources devoted to medical care, including the component absorbed by insurance. Out-of-pocket spending captures only the household’s direct cost share and is therefore a poor proxy for the biological input—the actual quantity of medical services received. Since insurance compresses the out-of-pocket distribution relative to the total spending distribution, using out-of-pocket spending would attenuate the estimated relationship between medical resources and survival.

An initial estimation using medical spending shares (out-of-pocket spending as a fraction of wealth) failed to identify  $\alpha$ . In that specification, the point estimate was  $\hat{\alpha} = 0.124$  with a robust standard error of 0.199 ( $t = 0.62$ ,  $p = 0.534$ ), and the 95% confidence interval included zero. The spending share variable lacked sufficient cross-sectional variation within the selected age group to separately identify the marginal product of medical spending from the baseline health parameter  $\underline{h}$ . Replacing the spending share with imputed total medical expenditure—which exhibits substantially greater cross-sectional dispersion—resolved this identification problem.

The estimation sample consists of single-person households in the HRS aged 55–64, pooled across waves 8–13 (2006–2016). Singles are used as the estimation sample because they serve as the empirical proxy for the high-utilization (low-health) state in the model. Among the elderly, nursing home residence is the closest empirical analogue to this state. Since single-person households—particularly widows and widowers—enter nursing homes at substantially higher rates than couples, the medical spending–survival relationship estimated on singles more accurately captures the health production function relevant to the low-health state. The pooled longitudinal sample yields 4,712 observations with non-missing imputed total medical expenditure and self-reported survival expectations.

Table 1 reports the estimation results.

Both parameters are precisely estimated. The estimate  $\hat{\alpha} = 0.730$  ( $t = 3.63$ ) confirms that total medical spending has a statistically significant effect on survival, validating the

**TABLE 1.** Nonlinear Least Squares Estimates of Health Production Function

| Parameter                                                  | Estimate | Robust S.E. | $t$    | $p$ -value | 95% C.I.       |
|------------------------------------------------------------|----------|-------------|--------|------------|----------------|
| $\alpha$                                                   | 0.730    | 0.201       | 3.63   | 0.000      | [0.336, 1.124] |
| $\beta$                                                    | 0.391    | 0.001       | 340.98 | 0.000      | [0.389, 0.393] |
| $N = 4,712, \quad R^2 = 0.997, \quad \underline{h} = 53.2$ |          |             |        |            |                |

*Notes: Nonlinear least squares estimation of  $\chi = 1 - (\underline{h} + \alpha \ln m)^{-\beta}$ , where  $\chi$  is self-reported probability of surviving to age 75 and  $m$  is imputed total medical expenditure. Sample: single-person HRS households aged 55–64, waves 8–13 (2006–2016). Robust standard errors.*

central mechanism of the model. The estimate  $\hat{\beta} = 0.391$  implies that the condition for health spending to act as a luxury good,  $\sigma > 1 + \beta$  (Theorem 3.2), is satisfied with  $\sigma = 2$ : since  $1 + 0.391 = 1.391 < 2$ , the curvature of marginal utility exceeds the curvature of the marginal product of health spending, ensuring that the precautionary saving motive does not vanish with wealth.

## Remaining Parameters

Calibration targets are  $\{\bar{y}, \underline{y}, \Gamma_y, \bar{h}, \underline{h}, \Gamma_h, \underline{m}, \underline{b}, \rho, \alpha, \beta\}$ . We set  $\bar{y}$  and  $\underline{y}$  to mean income for couple and single households, respectively, as singles are less wealthy than couples (De Nardi et al., 2025). High utilization is proxied by nursing-home residence. Because few individuals aged 55–65 live in nursing homes, we assume singles are more likely to enter this state. Income transitions are based on early retirement to construct  $\Gamma_y$ .

Retirement is modeled as absorbing. The mean retiree income is \$41,000. In HRS data, 293 of 1,717 employed households retire between waves, giving a 0.17 probability of moving to low income; average income of non-retirees is approximately \$80,000. The high-utilization (nursing home) state is also absorbing. Using spousal loss as a proxy, the probability of entering this state is 8% over two waves.

Parameters not identified from our data follow Hosseini (2015): borrowing rate 4%, risk aversion  $\sigma = 2$ , discount factor  $\rho = 0.97$ . We adopt  $\nu = 20$  from Hall and Jones (2007) for the value-of-statistical-life parameter, since the model excludes leisure. Larger values of  $\nu$  raise upper-tail saving only modestly.

**TABLE 2.** Calibration Summary

| Parameter | $\bar{y}$ | $\underline{y}$ | $\Gamma_y$                                           | $\bar{h}$ | $\underline{h}$ | $\Gamma_h$                                           | $\bar{m}$ | $\underline{b}$ | $\rho$     | $\alpha$ | $\beta$ |
|-----------|-----------|-----------------|------------------------------------------------------|-----------|-----------------|------------------------------------------------------|-----------|-----------------|------------|----------|---------|
| Value     | 80.71     | 41.67           | $\begin{bmatrix} 1 & 0 \\ 0.17 & 0.83 \end{bmatrix}$ | 9999      | 53.2            | $\begin{bmatrix} 1 & 0 \\ 0.08 & 0.92 \end{bmatrix}$ | 647       | -646.7          | 0.97       | 0.730    | 0.391   |
| Source    | HRS       | HRS             | HRS                                                  | by const. | NLS             | HRS                                                  | Medicaid  | $= \bar{m}$     | H&S (2015) | NLS      | NLS     |

Notes: All monetary values in thousands of dollars.  $\bar{y}$  and  $\underline{y}$  are mean income for couple and single households.  $\Gamma_y$  and  $\Gamma_h$  are transition matrices for income and health states.  $\bar{m}$  is the average long-term Medicaid payment.  $\underline{b}$  is the borrowing limit, set equal to the Medicaid transfer by construction.  $\alpha$  and  $\beta$  are estimated via nonlinear least squares (Table 1).  $\rho$  follows Hosseini (2015). “H&S (2015)” refers to Hosseini (2015). “by const.” indicates  $\bar{h}$  is set to ensure near-certain survival in the high-health state.

## Model Fit

Table 3 compares model-predicted savings with observed average savings by wealth decile for couple households in the HRS, for both the full model with health shocks and the BAH benchmark without them.

**TABLE 3.** Model vs. Observed Savings by Wealth Decile (Thousand Dollars)

| Wealth Decile | Full Model | BAH Model | HRS Data | Ratio |
|---------------|------------|-----------|----------|-------|
| 1             | 50.1       | 0.1       | 26.9     | 1.86  |
| 2             | 50.1       | -0.1      | 23.7     | 2.12  |
| 3             | 50.2       | -0.4      | 26.2     | 1.92  |
| 4             | 50.2       | -1.1      | 35.5     | 1.41  |
| 5             | 50.3       | -2.7      | 45.4     | 1.11  |
| 6             | 50.6       | -6.8      | 34.8     | 1.45  |
| 7             | 51.5       | -16.8     | 33.8     | 1.52  |
| 8             | 55.7       | -46.0     | 71.8     | 0.78  |
| 9             | 82.1       | -148.5    | 77.7     | 1.06  |
| 10            | 339.4      | -605.5    | 1,092.3  | 0.31  |

Notes: Full Model savings are computed from the calibrated model in (FUL). BAH Model refers to the benchmark specification without health shocks. HRS savings are average changes in net worth between waves 3 and 4 (1996–1998) for couple households aged 55–64. All values in thousands of real dollars.

The ratio column reports Full Model savings divided by observed savings.

The two models produce starkly different predictions. The full model generates a nearly flat saving profile through deciles 1–7, ranging from \$50,000 to \$52,000, before accelerating sharply in the upper deciles. The BAH model, by contrast, predicts dissaving that grows

monotonically with wealth, ranging from approximately zero at decile 1 to  $-\$605,500$  at decile 10. This divergence reflects the central theoretical mechanism: in the absence of health expenditure shocks, the standard consumption-smoothing motive causes wealthier households to draw down assets more aggressively. The health-shock channel reverses the sign of saving across the entire wealth distribution.

The fit of the full model is strongest in the upper deciles. At decile 9, the full model predicts  $\$82,100$  against  $\$77,700$  observed—a ratio of 1.06—while the BAH model predicts dissaving of  $-\$148,500$ , missing both the sign and the magnitude of observed saving. This comparison indicates that the precautionary mechanism begins to engage at roughly the correct wealth threshold and that the health-shock channel is quantitatively essential for reproducing the positive saving observed near the top of the distribution.

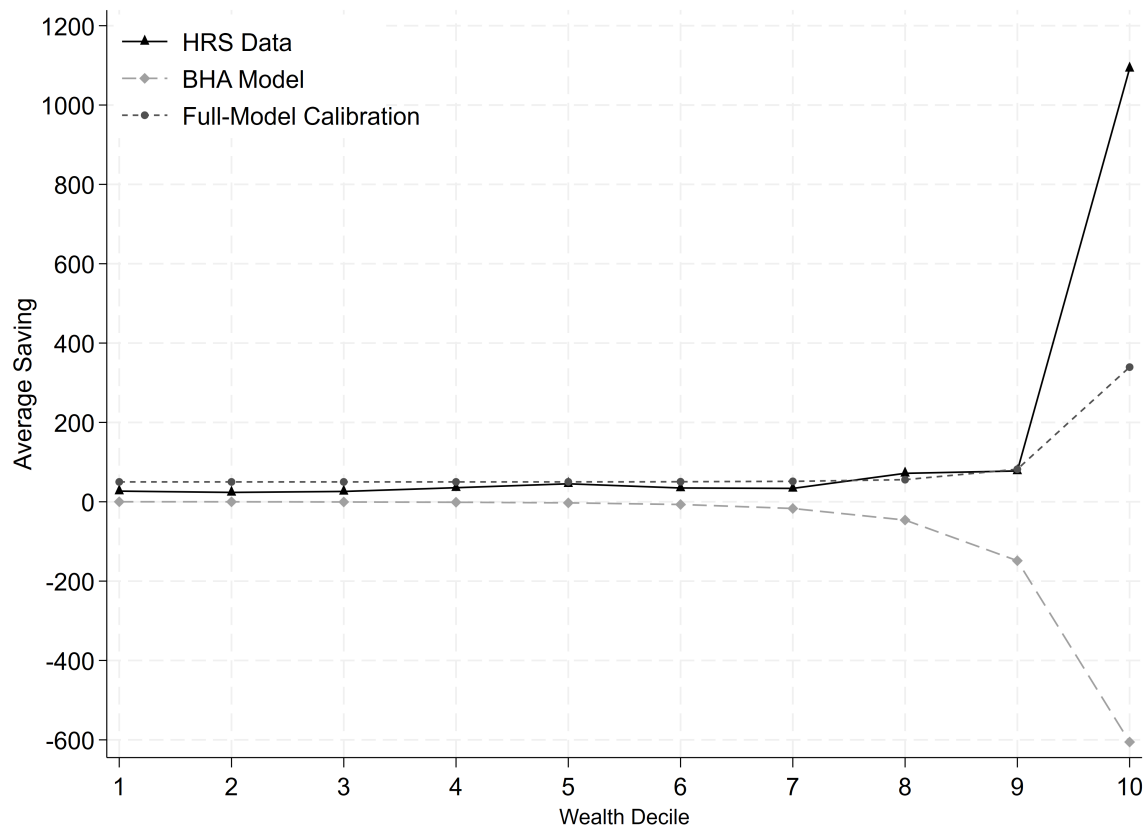
At the top decile, the full model predicts savings of  $\$339,400$  against  $\$1,092,300$  observed—a ratio of 0.31, indicating that the precautionary saving mechanism accounts for approximately 31 percent of observed average savings among the wealthiest couples. The BAH model predicts dissaving of  $-\$605,500$  at this same decile. The gap between the two model predictions—approximately  $\$945,000$ —measures the total contribution of the health-shock channel to saving at the top of the wealth distribution. This is quantitatively significant given that the full model contains a single novel channel—shocks to marginal continuation value—without invoking bequest motives, return heterogeneity, or entrepreneurial risk.

The full model overpredicts in deciles 1–7, where observed savings range from  $\$24,000$  to  $\$45,000$  but the model predicts approximately  $\$50,000$ . This is expected: the flat model prediction in this region reflects the near-constant baseline saving driven by the discount rate, interest rate, and income process, whereas the data exhibit non-monotonic variation across deciles that likely reflects heterogeneity in housing returns, pension accumulation, and other factors outside the model. The underprediction at decile 8 ( $\$55,700$  versus  $\$71,800$ ) suggests that the transition from the flat region to the accelerating region occurs slightly later in the model than in the data. In these lower deciles, the BAH model predicts saving levels closer to zero or modestly negative, undershooting the data by a wider margin than the full model overshoots it.

Figure 3 displays these results visually.

The residual gap between the full model and the data at the top decile—approximately

**FIGURE 3.** Calibration Fit



*Average savings of couple households across wealth deciles (HRS 1996–1998) compared with predictions from the full model and the BAH model. Values in thousands of dollars. The full model captures the acceleration of saving at the top of the wealth distribution, accounting for approximately 31 percent of observed savings in the top decile through the precautionary saving channel alone. The BAH model predicts monotonically increasing dissaving across deciles.*

\$753,000—is consistent with the presence of additional saving motives that the model deliberately omits, including bequest motives (De Nardi, 2004), return heterogeneity (Benhabib et al., 2015), and entrepreneurial saving (Cagetti and De Nardi, 2006). The contribution of our mechanism is that it identifies a distinct precautionary channel that operates even in the absence of these alternatives. Moreover, the correlation between saving rates and capital income rises with wealth while the correlation with earnings remains flat (see Appendix B.4.5), suggesting that return heterogeneity and precautionary motives may operate jointly at the top of the distribution.

## 6 Conclusion

This paper introduced a novel mechanism linking wealth accumulation to shocks in marginal continuation utility rather than income alone. The central theoretical result (Theorem 3.2) shows that when households face idiosyncratic longevity shocks that alter the continuation value of utility, and when the curvature of marginal utility exceeds the curvature of the marginal product of health spending ( $\sigma > 1 + \beta$ ), maintaining marginal utility becomes a luxury relative to consumption. Under this condition, the precautionary saving motive does not vanish with wealth—in contrast to the standard Bewley (1977); Huggett (1993); Aiyagari (1994) framework, where saving eventually becomes negative above a finite threshold. A further implication (Theorem 4.2) is that the combination of positive saving at the tail and random separation generates a stationary wealth distribution that follows Zipf’s law in the upper tail, providing a micro-founded explanation for the Pareto tail observed in U.S., Canadian, and European wealth data.

The calibration to HRS data on U.S. couples aged 55–65 confirms that the mechanism is quantitatively relevant. The model accounts for approximately 31 percent of observed average savings in the top wealth decile—a substantial share given that it operates through a single channel without invoking bequest motives, return heterogeneity, or entrepreneurial risk. The BAH benchmark without health shocks predicts monotonically increasing dis-saving across the wealth distribution, reaching  $-\$605,500$  at the top decile—missing both the sign and magnitude of observed saving. The health-shock channel thus not only increases the level of saving but reverses its sign, indicating that precautionary motives from

longevity-contingent health expenditure are quantitatively essential for generating the positive saving rates observed among wealthy older households. The model also reproduces the empirical acceleration of saving in the upper deciles and closely matches saving at the ninth decile, where the precautionary mechanism begins to engage.

The estimation of the health production function relied on imputed total medical expenditure (see Appendix C for the imputation procedure and its theoretical properties). This connection proved essential: an initial specification using out-of-pocket spending shares failed to identify the marginal product of medical spending ( $\alpha$ ), yielding a statistically insignificant estimate. Replacing spending shares with imputed total expenditure—which recovers the insurer-paid component that out-of-pocket measures miss—resolved this identification problem and produced a precisely estimated  $\hat{\alpha}$ . This illustrates a broader point: empirical work on the relationship between health spending and economic outcomes can be materially affected by the choice of spending measure, a finding consistent with the spending–wealth gradient documented in Appendix C.

The empirical patterns documented in Appendix B further support the mechanism proposed here. Among couples, saving increases monotonically with wealth and saving rates flatten rather than decline at the top of the distribution—patterns inconsistent with standard income-fluctuation models but consistent with a precautionary motive that remains active among the wealthy. Moreover, the correlation between saving rates and capital income rises across wealth deciles while the correlation with earnings remains flat, suggesting that precautionary motives and return heterogeneity may operate jointly at the top of the distribution.

Several limitations should be noted. First, the theoretical analysis maintains the assumption that the value function in the low-health state is concave (Assumption 3.1). This assumption cannot be verified analytically because the product of concave functions on the right-hand side of the functional equation does not preserve concavity under standard inheritance arguments. In all numerical experiments, the implied policy functions are increasing, which is consistent with concavity, but a formal proof remains an open question. Second, the parameters of the health production function are estimated on single-person households and applied to a couple model. This choice is motivated by the higher nursing home entry rates among singles, which make them a better empirical proxy for the high-utilization state,

but it introduces a potential mismatch between the estimation and simulation populations. Third, the model produces a nearly flat saving profile in the lower deciles, overpredicting observed savings by roughly \$15,000–\$25,000. This reflects the absence of mechanisms—such as housing returns, pension heterogeneity, and labor supply decisions—that generate cross-sectional variation in saving below the wealth threshold at which the precautionary channel activates. Fourth, the condition  $\sigma > 1 + \beta$  is sufficient but not necessary for the result; alternative parameterizations of the value of being alive ( $\nu$ ) or the health production function ( $\alpha$ ) may also sustain positive saving at the tail, as suggested by numerical experiments.

These findings have implications for several areas of policy. The result that precautionary saving against health shocks drives wealth accumulation at the top suggests that the design of public health insurance—particularly long-term care coverage and Medicaid asset limits—can have first-order effects on the wealth distribution. Means-testing of benefits, which conditions transfers on household wealth, may inadvertently discourage precautionary saving among households approaching the eligibility threshold while leaving the saving motive fully intact for those well above it. More generous public insurance in the low-health state would reduce the marginal value of private saving for survival, potentially compressing the upper tail of the wealth distribution. Quantifying these effects requires embedding the mechanism in a general-equilibrium framework with endogenous prices and government budget constraints, which is a natural direction for future work.

A second direction is to extend the model to allow for joint health and income shocks, richer household structure (including spousal health interactions and widowhood risk), and endogenous labor supply. The current framework treats income transitions as independent of health, an assumption that abstracts from the well-documented relationship between health deterioration and labor force exit. Relaxing this assumption would allow the model to speak to the interaction between disability insurance, retirement timing, and precautionary wealth accumulation. Finally, incorporating the model into a life-cycle framework with realistic mortality tables and Social Security benefits would permit a direct comparison with the structural estimation literature on elderly saving, medical spending, and long-term care (De Nardi et al., 2010; French and Kelly, 2016).

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# Appendices

## A Theoretical Appendix

### Huggett (1993): Theorem 2

**LEMMA A.1** For all  $b > \underline{b}$ ,  $b'_l(b) \leq b$ .

*Proof.* The Euler condition for the low-type with wealth  $b$  is:

$$V_b(b) = (1 + r) \rho \chi(\underline{m}(b)) V_b(b'_l(b))$$

Since  $(1 + r) \rho \chi(\underline{m}(b)) < 1$ :

$$V_b(b) < V_b(b'_l(b))$$

which under assumption 3.1 implies  $b'_l(b) < b$ . □

Note that  $b'_l(-\underline{b}) \leq -\underline{b}$  can only hold as a strict equality. As a result,  $b'_l(b)$  must pass through the point  $(-\underline{b}, -\underline{b})$  and lie under the 45-degree line for all values of  $b$ .

**LEMMA A.2** When  $b'_h(b) > b$ , then  $c_h(b) - c_l(b) < e_h - e_l$ .

*Proof.* By lemma A.1, we know that  $rb + e_l - c_l(b) \leq 0$ , or

$$c_l(b) - e_l \geq rb.$$

On the other hand, for  $b'_h(b) > b$ , we need to have  $c_h(b) - e_h < rb$ , or

$$-[c_h(b) - e_h] > -rb.$$

If we add these two inequalities, we get the intended result. □

A corollary of this is that

$$0 < b'_h(b) - b'_l(b) = e_h - e_l - [c_h(b) - c_l(b)] < e_h - e_l. \quad (15)$$

Therefore,

$$b'_l(b) \geq b - (e_h - e_l). \quad (16)$$

If we let  $\kappa := (e_h - e_l)$ , from lemma A.2, we can write  $c_h(b) - \kappa < c_l(b) \leq c_h(b)$ , where the second inequality follows from the monotonicity of optimal consumption. Therefore,

$$u'(c_l(b)) < u'(c_h(b) - \kappa) \approx u'(c_h(b)) - \kappa u''(c_h(b)). \quad (17)$$

Let us plug this into the Euler equation for the high-types:

$$\begin{aligned} u'(c_h(b)) &= \frac{\rho}{(1+r)} [\pi_h u'(c_h(b'_h(b))) + (1 - \pi_h) u'(c_l(b'_h(b)))] \\ &< \pi_h u'(c_h(b'_h(b))) + (1 - \pi_h) u'(c_l(b'_h(b))) \\ &< \pi_h u'(c_h(b'_h(b))) + (1 - \pi_h) [u'(c_h(b'_h(b))) - \kappa u''(c_h(b'_h(b)))] . \end{aligned} \quad (18)$$

The last inequality can be rearranged and written as

$$u'(c_h(b)) - u'(c_h(b'_h(b))) < -(1 - \pi_h) \kappa u''(c_h(b'_h(b))). \quad (19)$$

If we replace for the first and second derivatives of the utility function, assuming a CRRA felicity, we have

$$\frac{1}{[c_h(b)]^\sigma} - \frac{1}{[c_h(b'_h(b))]^\sigma} < \frac{(1 - \pi_h) \kappa}{[c_h(b'_h(b))]^{\sigma+1}}. \quad (20)$$

The next theorem uses this inequality to show that the individual's saving rate must eventually become negative, regardless of her level of current income.

**THEOREM A.3** *There must exist some  $\underline{b} < \bar{b} < \infty$  for which, for all  $b \geq \bar{b}$ , we must have  $rb + e_h - c_h(b) \leq 0$ .*

*Proof.* Multiplying both sides of (20) by  $[c_h(b'_h(b))]^\sigma$ :

$$\frac{[c_h(b'_h(b))]^\sigma}{[c_h(b)]^\sigma} - 1 < \frac{(1 - \pi_h)\kappa}{c_h(b'_h(b))} \quad (21)$$

Rearranging:

$$\left[ \frac{c_h(b'_h(b))}{c_h(b)} \right]^\sigma < 1 + \frac{(1 - \pi_h)\kappa}{c_h(b'_h(b))} \quad (22)$$

Taking the limit as  $b \rightarrow \infty$ , and noting that  $c_h(b) \rightarrow \infty$  and  $c_h(b'_h(b)) \rightarrow \infty$ , the right-hand side converges to 1, so in the limit:

$$\left[ \frac{c_h(b'_h(b))}{c_h(b)} \right]^\sigma \leq 1 \quad (23)$$

which implies  $c_h(b'_h(b)) \leq c_h(b)$ . Since  $c_h$  is an increasing function, this implies:

$$b'_h(b) \leq b \quad (24)$$

That is, in the limit for  $b \geq \bar{b}$  we have  $b'_h(b) - b = rb + e_h - c_h(b) \leq 0$   $\square$

## B Saving Patterns across the Wealth Distribution

### B.1 Introduction

This appendix examines the distribution and sources of saving. It describes the Health and Retirement Study (HRS) and discusses cohort selection based on the research question. It also evaluates the external validity of the selected sample by benchmarking its wealth distribution against Federal Reserve (FRED) data.

The appendix then reviews alternative definitions of saving, identifies the definition best suited for studying the saving distribution, and defines corresponding saving rates. It also discusses practical challenges and potential measurement issues when implementing these definitions in the data.

Next, the appendix studies the distributional properties of wealth, saving, and saving rates.

It characterizes dispersion and asymmetry in each distribution and compares these patterns with those documented in administrative data used in the literature. It further examines how these distributions differ between single and couple households.

It then reports the relationship between saving and wealth by comparing saving levels and saving rates across wealth deciles. In contrast to standard theoretical predictions, saving increases with wealth, while saving rates flatten in higher deciles.

Wealth returns help explain heterogeneity in saving across the distribution. In particular, the correlation between saving and wealth returns exhibits a clear upward pattern across deciles, rising in higher deciles. This suggests that wealth returns are an important driver of saving among households at the top of the wealth distribution.

## B.2 Definitions and Measurement

This section defines the saving measures used throughout the appendix and discusses practical considerations that arise when implementing them in the data. We define three objects: total saving, active saving, and saving rates. Total saving measures the change in household net worth between survey waves. Active saving removes the component of wealth growth attributable to returns on existing assets, isolating the portion that reflects discretionary accumulation. Saving rates scale each measure by net worth, allowing comparisons across households with different wealth levels.

### Total Saving

The primary measure of saving is the change in household net worth between consecutive survey waves:

$$S_{it} = b_{it+1} - b_{it} \tag{25}$$

where  $b_{it}$  denotes the net worth of household  $i$  at the end of period  $t$ . Since the HRS interviews households every two years,  $S_{it}$  captures wealth accumulation over a two-year horizon.

An alternative derivation proceeds from the household budget constraint. Let  $r$  denote the return on wealth,  $y_{it}$  labor income,  $c_{it}$  consumption, and  $m_{it}$  medical and other non-

consumption expenditures. The budget constraint implies:

$$b_{it+1} = b_{it} + r b_{it} + y_{it} - c_{it} - m_{it} \quad (26)$$

so that saving can equivalently be expressed as the difference between total income and total expenditure:

$$S_{it} = r b_{it} + y_{it} - c_{it} - m_{it} \quad (27)$$

Although the two expressions are equivalent in theory, they differ in practice because each relies on different variables measured with different degrees of precision. The income-expenditure definition requires accurate measurement of consumption, income, and returns—each of which is subject to substantial measurement error in survey data (Juster et al., 1999). The first-differencing definition in (25) requires only two observations of net worth. Since the HRS collects detailed asset and liability data at the household level, we adopt (25) as our baseline measure.

### Active Saving

Total saving includes the component of wealth growth generated by returns on existing assets. In many cases—particularly for financial wealth held in retirement accounts, index funds, or other managed portfolios—these returns are automatically reinvested without any discretionary action by the household. To isolate the portion of saving that reflects active household decisions, we define active saving as:

$$S_{it}^{act} = S_{it}^{tot} - r_{it} b_{it} \quad (28)$$

where  $r_{it} b_{it}$  denotes capital income. The distinction between total and active saving is informative about the sources of wealth accumulation. If total and active saving rates diverge substantially in a given wealth bracket, this indicates that returns on existing wealth—rather than new saving out of income—account for most of the observed wealth growth. As we document in Section B.4, this divergence is particularly pronounced at the top of the wealth distribution.

## Saving Rates

To compare saving intensity across households with different wealth levels, we define the total saving rate and the active saving rate as:

$$s_{it}^{tot} = \frac{S_{it}^{tot}}{b_{it}}, \quad s_{it}^{act} = \frac{S_{it}^{act}}{b_{it}} \quad (29)$$

This normalization introduces several measurement issues. First, households with zero net worth produce an undefined saving rate. We exclude these observations: 56 couple households and 539 single households. Second, households with negative net worth are retained. For these households, the saving rate measures the share of debt repaid; for example, a household that reduces its net liabilities from \$100,000 to \$50,000 has a saving rate of one-half. Third, because the HRS measures wealth every two years, the saving rate reflects a two-year accumulation rate rather than an annual flow. We report all saving rates on a per-wave (two-year) basis throughout.

A further consideration is the treatment of outliers. Both saving levels and saving rates exhibit extreme values at the tails of the wealth distribution, driven by large capital gains or losses, inheritances, or measurement error in asset valuations. Rather than trimming observations *ex ante*, we report robust measures of dispersion—the interdecile range alongside the standard deviation—and quantile-based skewness to characterize the shape of the distribution without being dominated by extreme observations. This approach follows the methodology used in recent studies of saving behavior using administrative data (Bach et al., 2017; Fagereng et al., 2019).

## B.3 Data and Sample Selection

This section describes the data, explains the cohort and age-group selection, documents the sample restrictions, and benchmarks the wealth distribution in the selected sample against Federal Reserve data.

## **Health and Retirement Study**

The Health and Retirement Study (HRS) is a longitudinal panel survey of Americans over the age of 50, sponsored by the National Institute on Aging (grant number NIA U01AG009740) and conducted by the University of Michigan. Households are interviewed every two years, beginning in 1992, with asset and liability data collected at the household level at the end of each survey year. We use the RAND HRS Longitudinal File, a cleaned and harmonized version of the data that includes derived variables for demographics, income, and wealth components.

The HRS follows several cohorts, each entering the sample at a different date and re-interviewed biennially thereafter. To maintain representativeness of the aging population, new cohorts are added periodically. Since wealth is recorded at the household level, the unit of analysis differs by household type: for singles, the individual and the household coincide; for couples, both spouses share a single wealth observation.

## **Cohort and Age-Group Selection**

The analysis focuses on households in wave 3 (the 1996 interview) with at least one member aged 55 to 64. This choice is motivated by two considerations. First, this age-wave cell contains the largest number of observed couple households—1,555 couples along with 1,610 singles—ensuring adequate sample size for decile-level analysis. Table 4 reports the number of households across age groups and waves for both household types, confirming that the 55–64 group in wave 3 provides the best combination of sample size and demographic homogeneity.

Second, the 55–64 age range captures households in the final decade before typical retirement. These households face active portfolio and saving decisions while simultaneously confronting rising health risk and approaching Social Security eligibility. This makes them well suited for studying the relationship between wealth accumulation and saving behavior that motivates the main text.

Since the HRS interviews households every two years, saving is measured as the change in net worth between wave 3 (1996) and wave 4 (1998), yielding a two-year saving flow for each household.

**TABLE 4.** Number of Households by Age Group and Wave

| a. Couples |              |        |       |       | b. Singles |              |        |        |       |
|------------|--------------|--------|-------|-------|------------|--------------|--------|--------|-------|
| Wave       | 55–64        | 65–74  | 75–84 | 85–94 | Wave       | 55–64        | 65–74  | 75–84  | 85–94 |
| 1992       | 1,468        | —      | —     | —     | 1992       | 1,341        | 7      | —      | —     |
| 1994       | 1,507        | 741    | 485   | 70    | 1994       | 1,480        | 1,077  | 1,896  | 690   |
| 1996       | <b>1,555</b> | 582    | 504   | 79    | 1996       | <b>1,610</b> | 705    | 1,886  | 876   |
| 1998       | 1,552        | 1,171  | 507   | 77    | 1998       | 1,402        | 1,365  | 1,778  | 989   |
| 2000       | 1,416        | 1,124  | 498   | 64    | 2000       | 1,122        | 1,277  | 1,528  | 961   |
| 2002       | 1,265        | 1,142  | 552   | 61    | 2002       | 936          | 1,271  | 1,318  | 925   |
| 2004       | 1,091        | 1,120  | 564   | 74    | 2004       | 858          | 1,194  | 1,071  | 838   |
| 2006       | 880          | 1,115  | 573   | 81    | 2006       | 772          | 1,183  | 897    | 774   |
| 2008       | 833          | 1,107  | 615   | 93    | 2008       | 815          | 1,018  | 807    | 716   |
| 2010       | 1,125        | 1,017  | 599   | 89    | 2010       | 1,685        | 818    | 824    | 617   |
| 2012       | 1,213        | 917    | 612   | 114   | 2012       | 1,823        | 674    | 792    | 478   |
| 2014       | 1,176        | 791    | 570   | 107   | 2014       | 1,839        | 692    | 763    | 392   |
| 2016       | 1,170        | 664    | 542   | 106   | 2016       | 2,180        | 822    | 719    | 346   |
| 2018       | 944          | 613    | 510   | 94    | 2018       | 1,961        | 839    | 544    | 284   |
| Total      | 17,195       | 12,104 | 7,131 | 1,109 | Total      | 19,824       | 12,942 | 14,823 | 8,886 |

*Notes: This table reports the number of households by age group and interview wave. Panel (a) shows couple households; panel (b) shows single-person households. The selected analysis sample—wave 3, ages 55–64—is indicated in bold. Cells marked “—” indicate that the cohort had not yet entered the sample.*

### Sample Restrictions

We impose several restrictions on the analysis sample. First, we require that saving rates be non-missing. Since the saving rate is defined as saving divided by net worth, this restriction excludes two types of households: those that attrit between waves 3 and 4 (and therefore lack the wealth observation needed to compute saving), and those with exactly zero net worth (for which the saving rate is undefined). For couple households, this condition must hold for both spouses; if either spouse has a missing saving rate, the entire household is dropped. This restriction removes 181 couple households and 375 single households.

Second, we exclude households with extreme saving rates. Observations with saving rates below the 1st percentile or above the 99th percentile of the saving rate distribution are dropped. For couple households, if either spouse’s saving rate falls in the extreme tails,

both spouses are excluded. This trimming removes an additional 26 couple households and 22 single households, limiting the influence of outliers arising from large capital gains or losses, inheritances, or measurement error in asset valuations.

Households with negative net worth are retained throughout. For these observations, the saving rate measures the share of debt repaid between waves.

After all restrictions, the analysis sample comprises 1,346 couple households and 1,171 single-person households.

### **Wealth Concentration in the Sample**

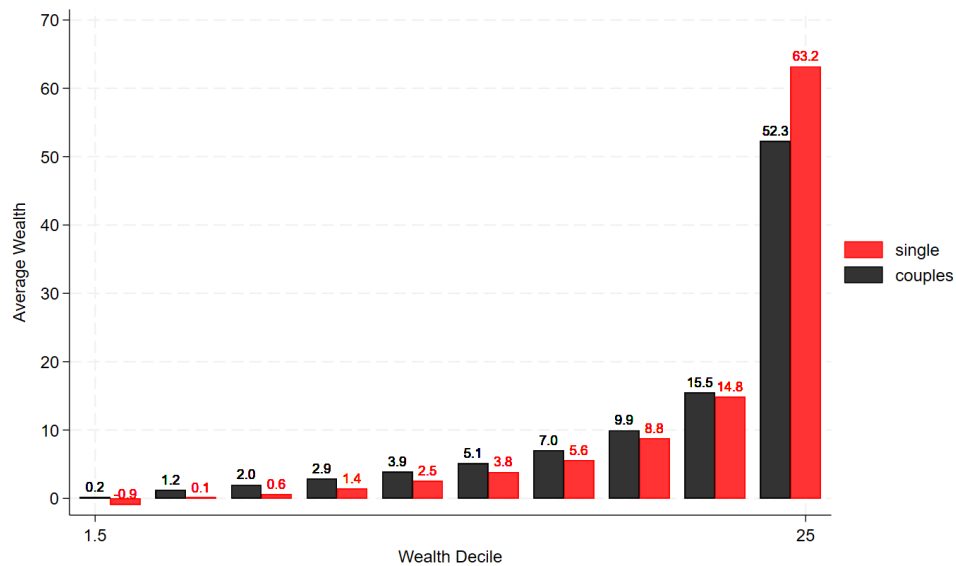
Figure 4 displays the share of total wealth held by each wealth decile in our sample, separately for couples and singles. The wealth distribution is heavily concentrated at the top: among singles, the top decile holds 63.2 percent of total wealth, while among couples the corresponding share is 52.2 percent. This degree of concentration is broadly consistent with the patterns documented in aggregate U.S. wealth data, where the top 10 percent of the distribution typically accounts for the majority of total net worth.

## **B.4 Results**

This section presents the main empirical findings. It begins with the distribution and composition of wealth, then examines saving levels and saving rates across wealth deciles. It characterizes the dispersion and symmetry of saving rates and compares these patterns with those documented in administrative data from other countries. Finally, it examines the correlation between saving rates and different components of income, highlighting the role of wealth returns.

Throughout, we analyze couple and single households separately. As discussed in the previous section, wealth is measured at the household level, so the two groups are not directly comparable in levels. The focus is on how saving behavior varies across the wealth distribution within each group.

**FIGURE 4.** Share of Total Wealth by Decile



*Notes: This figure displays the share of total wealth held by each wealth decile among households aged 55–64 in wave 3 (1996). Red bars indicate single-person households; black bars indicate couples. The share is computed as the sum of net worth within each decile divided by the total net worth of the respective household type.*

### B.4.1 Wealth Distribution and Composition

Table 5 reports the mean and median net worth for each wealth decile, separately for singles and couples. For singles, the mean and median are closely aligned in the middle of the distribution, indicating relatively little within-decile skewness. The exceptions are the tails: the bottom decile has a mean net liability of \$21,000 against a median of \$3,000, reflecting a small number of heavily indebted households, while the top decile has a mean of \$1,442,000 against a median of \$833,000, indicating substantial right skew among the wealthiest singles.

**TABLE 5.** Mean and Median Wealth by Decile (Thousand Dollars)

| Wealth Decile | Singles |        | Couples |        |
|---------------|---------|--------|---------|--------|
|               | Mean    | Median | Mean    | Median |
| P0–P10        | –21     | –3     | 12      | 19     |
| P10–P20       | 2       | 1      | 72      | 72     |
| P20–P30       | 13      | 13     | 116     | 115    |
| P30–P40       | 32      | 32     | 168     | 169    |
| P40–P50       | 57      | 57     | 226     | 224    |
| P50–P60       | 87      | 87     | 299     | 299    |
| P60–P70       | 127     | 126    | 413     | 410    |
| P70–P80       | 197     | 194    | 580     | 565    |
| P80–P90       | 336     | 332    | 903     | 871    |
| P90–P100      | 1,442   | 833    | 3,070   | 1,820  |

*Notes: Net worth at the beginning of wave 3 (1996), in thousands of real dollars. The gap between mean and median within a decile indicates within-decile skewness.*

Couples hold substantially more wealth than singles at every point of the distribution. Even after dividing couple wealth by two to approximate a per-capita measure, the gap persists: at the bottom, per-capita couple wealth is \$6,000 compared to –\$21,000 for singles; at the top, it is \$1,535,000 compared to \$1,442,000. This difference likely reflects both selection into marriage and the economic advantages of resource pooling. Among couples, within-decile skewness is modest below the ninth decile but pronounced at the top, where the mean-median gap exceeds \$1.2 million.

Table 6 reports the composition of wealth across deciles. For singles, housing is the dominant asset from the third through the eighth decile, peaking at 68 percent of net worth in the fifth decile. Financial assets gain importance higher in the distribution, rising from –8 per-

cent in the third decile to 36 percent in the top decile, where they become the largest single component. The bottom two deciles are distinct: the first decile holds wealth almost entirely in financial assets (104 percent of net worth, offset by liabilities in other categories), while the second decile is concentrated in transportation and financial assets, consistent with households that do not own homes.

The composition of couple wealth follows a similar gradient but with greater diversification at higher wealth levels. Housing dominates for lower deciles, peaking at 69 percent in the second decile. As wealth increases, the portfolio shifts toward financial assets, pensions, real estate, and business equity, which collectively account for 83 percent of net worth in the top decile. Notably, business equity rises from under 3 percent in the bottom half to 17 percent at the top—a pattern largely absent among singles. Across all households, housing accounts for 42 percent of total wealth for both singles and couples. The key difference is that singles concentrate the remainder in financial and transportation assets, while couples hold a more diversified portfolio across pensions, real estate, and businesses.

This compositional shift is important for interpreting the saving results that follow. As documented below, wealth returns are more strongly correlated with saving at the top of the distribution, precisely where portfolios tilt toward return-generating assets such as financial holdings, real estate, and business equity.

#### **B.4.2 Saving Levels**

Figure 5 presents average saving—the change in net worth between waves 3 and 4—by wealth decile for both household types. The patterns differ sharply. Among couples, saving is positive throughout the distribution, ranging from \$23,000 to \$45,000 in the bottom seven deciles, rising to \$71,000 and \$78,000 in the eighth and ninth, and reaching \$1,092,000 in the top decile. The increase is monotonic and accelerates at the top.

Among singles, the pattern is different. The bottom decile saves \$21,000 on average, and the middle deciles save modest amounts between zero and \$15,000. The top decile, however, dissaves \$260,000 on average. This dissaving at the top is driven by large negative outliers—a feature we return to in the dispersion analysis below.

The divergence between couples and singles at the top of the wealth distribution is strik-

**TABLE 6.** Wealth Composition by Decile (Percent of Net Worth)*a. Singles*

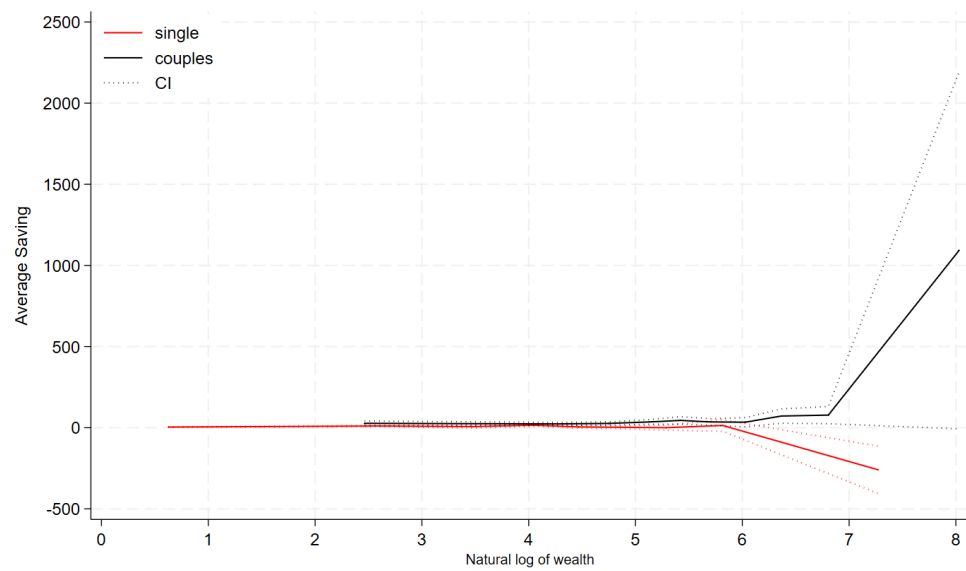
| Wealth Decile | Housing | Financial | Transport. | Pension | Real Estate | Business |
|---------------|---------|-----------|------------|---------|-------------|----------|
| P0-P10        | 1       | 104       | -4         | -1      | 0           | 0        |
| P10-P20       | 2       | 32        | 67         | 0       | 0           | 0        |
| P20-P30       | 52      | -8        | 43         | 7       | 5           | 1        |
| P30-P40       | 64      | 8         | 16         | 7       | 4           | 1        |
| P40-P50       | 68      | 10        | 13         | 4       | 4           | 1        |
| P50-P60       | 56      | 15        | 8          | 8       | 10          | 2        |
| P60-P70       | 60      | 14        | 9          | 8       | 9           | 1        |
| P70-P80       | 51      | 21        | 6          | 11      | 8           | 2        |
| P80-P90       | 38      | 28        | 4          | 13      | 14          | 3        |
| P90-P100      | 23      | 36        | 2          | 13      | 19          | 6        |
| Total         | 42      | 26        | 16         | 7       | 7           | 2        |

*b. Couples*

| Wealth Decile | Housing | Financial | Transport. | Pension | Real Estate | Business |
|---------------|---------|-----------|------------|---------|-------------|----------|
| P0-P10        | 39      | 32        | 32         | 1       | -3          | -1       |
| P10-P20       | 69      | 3         | 16         | 6       | 4           | 2        |
| P20-P30       | 68      | 9         | 13         | 3       | 5           | 1        |
| P30-P40       | 53      | 15        | 11         | 9       | 9           | 3        |
| P40-P50       | 49      | 14        | 11         | 13      | 9           | 3        |
| P50-P60       | 43      | 21        | 9          | 14      | 9           | 5        |
| P60-P70       | 34      | 22        | 7          | 18      | 15          | 4        |
| P70-P80       | 28      | 27        | 5          | 19      | 14          | 7        |
| P80-P90       | 23      | 27        | 4          | 19      | 19          | 9        |
| P90-P100      | 14      | 27        | 2          | 12      | 27          | 17       |
| Total         | 42      | 20        | 11         | 11      | 11          | 5        |

*Notes: Each cell reports the average share of net worth held in the given asset category, computed at the household level and averaged within each decile. Housing refers to the net value of the primary residence. Financial includes stocks, mutual funds, money market accounts, CDs, bonds, and other financial assets. Transport. refers to vehicles, boats, and similar assets. Pension denotes IRA and Keogh accounts. Real Estate is the net value of non-primary properties. Business is the net value of owned enterprises. Shares can exceed 100% or be negative when a household holds positive assets in one category but has negative net worth overall.*

**FIGURE 5. Average Saving by Wealth Decile**



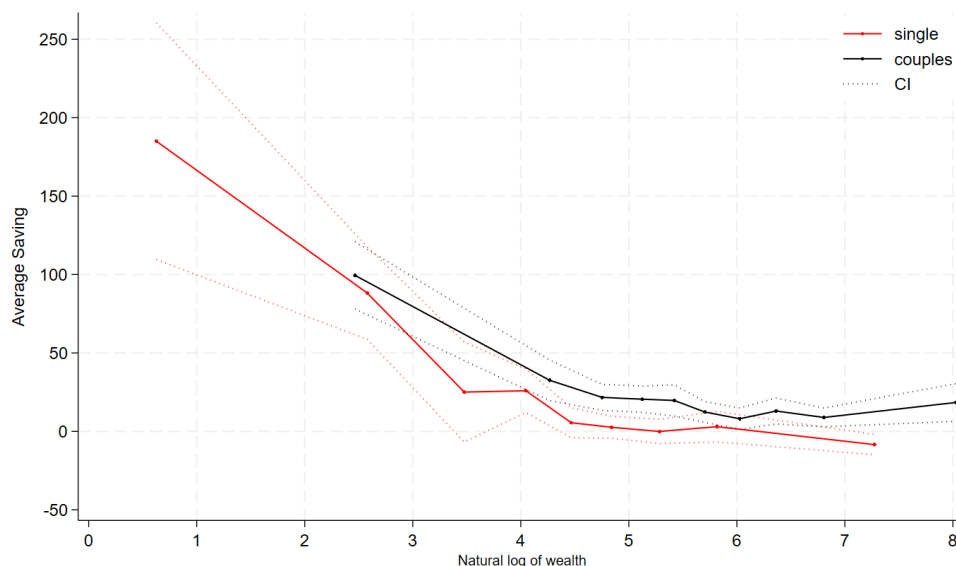
*Notes: This figure reports average saving (change in net worth from wave 3 to wave 4, in thousands of real dollars) by wealth decile. Gray circles indicate single-person households; black diamonds indicate couples. Wealth deciles are constructed separately for each household type.*

ing. In the highest decile, couples save \$1,092,000 while singles dissave \$260,000—a gap of \$1,352,000. This pattern is inconsistent with standard life-cycle models, which predict that wealthy households should draw down assets to finance consumption. While consumption data are not observed in the HRS, the magnitude of couple saving at the top suggests that these households maintain high consumption levels while simultaneously accumulating wealth—a pattern that requires explanation beyond standard consumption-smoothing motives.

### B.4.3 Saving Rates

To compare saving intensity across households with different wealth levels, Figure 6 reports saving rates—saving divided by net worth—by wealth decile. Among singles, the bottom two deciles exhibit elevated saving rates (172 and 185 percent), reflecting the small denominators in these groups. From the third decile onward, the saving rate declines steadily, reaching approximately zero by the eighth decile and turning negative (−8 percent) in the top decile.

**FIGURE 6.** Average Saving Rate by Wealth Decile



*Notes: This figure reports the average saving rate (saving divided by net worth) by wealth decile. Gray circles indicate singles; black diamonds indicate couples. Saving rates are computed at the household level over a two-year horizon (wave 3 to wave 4).*

For couples, the saving rate in the bottom decile is 99 percent, declining to 32 percent in the second and stabilizing around 20 percent in deciles three through five. It falls further to approximately 10 percent in deciles six through nine, then rises to 20 percent in the top decile. This U-shaped pattern at the top—where saving rates increase rather than continue declining—is a central finding of this appendix. Standard income-fluctuation models predict that saving rates become negative as wealth increases, since wealthier households can smooth consumption without additional accumulation. The data contradict this prediction for couples.

Table 7 decomposes the saving rate into contributions from each asset category. For singles, saving in the bottom deciles is driven by gains in housing, financial, and transportation assets. In the upper deciles, the pattern reverses: housing and real estate values decline as a share of net worth, while financial saving provides the only positive contribution—except in the top decile, where all major categories show declines.

For couples, financial and pension saving are the primary sources of wealth accumulation across most of the distribution. Unlike singles, couples do not exhibit dissaving in housing at the top. Business equity begins to contribute meaningfully from the seventh decile onward. Overall, couples tend to save across multiple asset classes rather than drawing down in one to fund accumulation in another—a pattern consistent with their more diversified portfolios.

Figure 7 compares total and active saving rates across wealth deciles. Active saving—which subtracts capital income from total saving—isolates the discretionary component of wealth accumulation. The two measures track each other closely in the lower deciles but diverge at the top, particularly for couples. In the top couple decile, the active saving rate is approximately 10 percentage points below the total rate, indicating that roughly half of observed saving at the top is attributable to returns on existing wealth rather than new accumulation out of income.

For singles, the gap between total and active saving is largest in the bottom two deciles—approximately 50 and 78 percentage points—consistent with the finding that financial saving drives wealth growth at the bottom. At the top, the two measures converge, reflecting the negative total saving rate in the top single decile.

The divergence between total and active saving rates at the top of the couple distribution underscores the importance of wealth returns for understanding saving behavior among the

**TABLE 7.** Saving Rate Composition by Decile (Percent of Net Worth)*a. Singles*

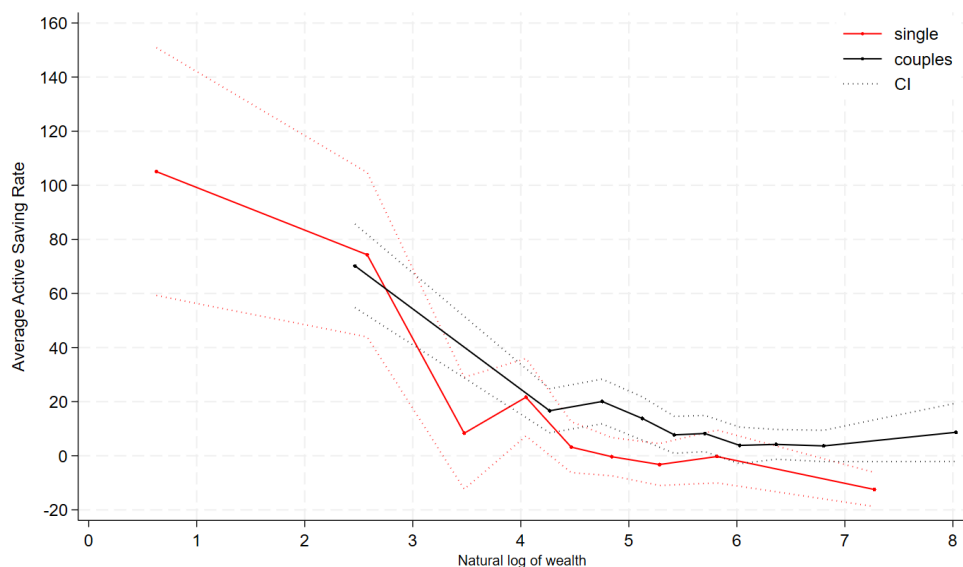
| Wealth Decile | Housing | Financial | Transport. | Pension | Real Est. | Business | Total |
|---------------|---------|-----------|------------|---------|-----------|----------|-------|
| P0–P10        | 37      | 87        | 42         | 6       | 0         | 0        | 172   |
| P10–P20       | 50      | 50        | 84         | 0       | 0         | 0        | 185   |
| P20–P30       | 76      | 14        | –2         | 0       | –3        | 4        | 89    |
| P30–P40       | 13      | 1         | –2         | 2       | 13        | –1       | 26    |
| P40–P50       | 18      | 4         | 2          | 1       | –1        | 1        | 25    |
| P50–P60       | –1      | 4         | 2          | 0       | –2        | 3        | 6     |
| P60–P70       | 0       | 3         | –1         | 7       | –5        | 0        | 4     |
| P70–P80       | –3      | 4         | 0          | 1       | –2        | 0        | –1    |
| P80–P90       | –2      | 5         | 1          | 1       | –5        | 3        | 3     |
| P90–P100      | –2      | –4        | 1          | 1       | –3        | –1       | –8    |
| Total         | 19      | 17        | 13         | 2       | –1        | 1        | 50    |

*b. Couples*

| Wealth Decile | Housing | Financial | Transport. | Pension | Real Est. | Business | Total |
|---------------|---------|-----------|------------|---------|-----------|----------|-------|
| P0–P10        | 42      | 40        | 18         | 1       | 0         | –1       | 100   |
| P10–P20       | 7       | 11        | 1          | 11      | 4         | 0        | 32    |
| P20–P30       | 1       | 9         | 2          | 3       | 2         | 4        | 22    |
| P30–P40       | 5       | 8         | 1          | 4       | 1         | –1       | 20    |
| P40–P50       | 1       | 16        | –1         | 8       | –3        | 0        | 20    |
| P50–P60       | 2       | 5         | 0          | 8       | –2        | 0        | 12    |
| P60–P70       | 0       | 4         | 0          | 6       | –3        | 1        | 8     |
| P70–P80       | 0       | 6         | 0          | 6       | –2        | 4        | 13    |
| P80–P90       | 0       | 3         | 1          | 7       | –2        | 1        | 9     |
| P90–P100      | 3       | 10        | 0          | 3       | –1        | 3        | 18    |
| Total         | 6       | 11        | 2          | 6       | –1        | 1        | 26    |

*Notes: Each cell reports the average ratio of saving in the given asset category to total net worth, computed at the household level and averaged within each decile. The Total column equals the sum of all components and corresponds to the total saving rate. All figures are in percent.*

**FIGURE 7.** Total vs. Active Saving Rates by Wealth Decile



*Notes: This figure compares total saving rates (solid lines) with active saving rates (dashed lines) by wealth decile. Active saving subtracts capital income from total saving. The horizontal line at zero is included for reference.*

wealthy. We examine this relationship more formally below.

#### B.4.4 Dispersion and Symmetry

Table 8 reports three measures of the saving rate distribution within each wealth decile: the standard deviation, the interdecile range (the difference between the 90th and 10th percentiles), and quantile-based skewness.<sup>10</sup> The interdecile range provides a robust complement to the standard deviation: in a standard normal distribution, the ratio of the interdecile range to the standard deviation is approximately 2.56, so ratios substantially below this threshold suggest the presence of heavy tails or outliers that inflate the standard deviation relative to the interquartile spread.

<sup>10</sup>Quantile-based skewness is defined as  $(P_{90} + P_{10} - 2 \cdot P_{50}) / (P_{90} - P_{10})$ , where  $P_k$  denotes the  $k$ th percentile. This measure is bounded between  $-1$  and  $1$  and is robust to outliers.

**TABLE 8.** Dispersion and Symmetry of Saving Rates

|          | Standard Deviation |         | Interdecile Range |         | Skewness |         |
|----------|--------------------|---------|-------------------|---------|----------|---------|
|          | Singles            | Couples | Singles           | Couples | Singles  | Couples |
| P0–P10   | 430%               | 174%    | 641%              | 448%    | 0.38     | 0.47    |
| P10–P20  | 610                | 104     | 886               | 176     | 0.80     | 0.35    |
| P20–P30  | 239                | 68      | 521               | 133     | 0.55     | 0.19    |
| P30–P40  | 258                | 67      | 238               | 139     | 0.17     | 0.28    |
| P40–P50  | 112                | 81      | 192               | 139     | 0.30     | 0.22    |
| P50–P60  | 76                 | 54      | 171               | 111     | −0.02    | 0.16    |
| P60–P70  | 57                 | 55      | 134               | 124     | 0.03     | 0.29    |
| P70–P80  | 62                 | 67      | 132               | 117     | 0.17     | 0.19    |
| P80–P90  | 79                 | 48      | 155               | 111     | 0.22     | 0.12    |
| P90–P100 | 52                 | 97      | 136               | 150     | 0.23     | 0.26    |
| Total    | 198                | 82      | 320               | 165     | 0.28     | 0.25    |

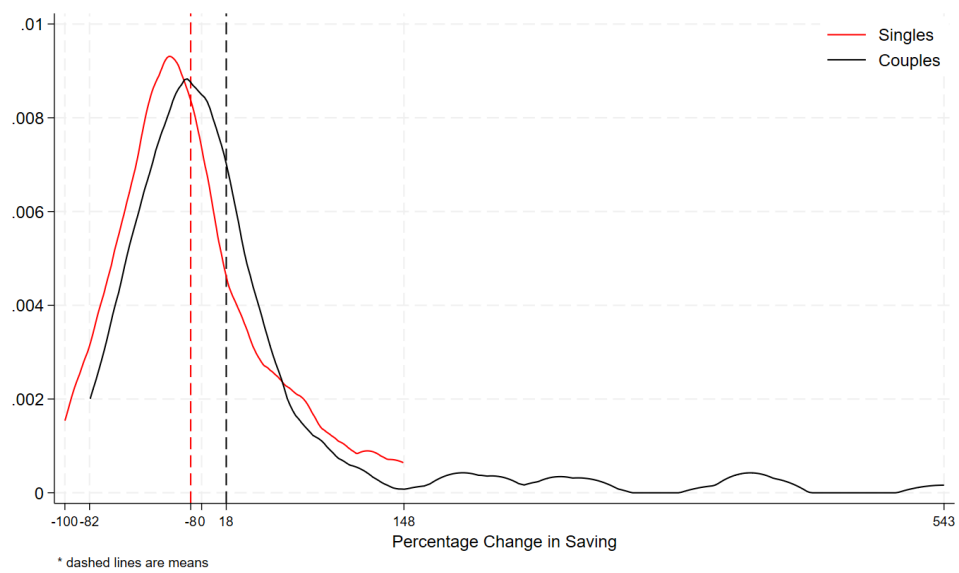
Notes: Standard deviation and interdecile range (P90 – P10) are expressed in percentage points. Skewness is quantile-based:  $(P_{90} + P_{10} - 2 \cdot P_{50}) / (P_{90} - P_{10})$ .

Two patterns emerge. First, dispersion declines with wealth for both household types, but much more steeply for singles. The standard deviation of the single saving rate falls from 430 percent in the bottom decile to 52 percent at the top—an eightfold reduction. For couples, the decline is more moderate, from 174 to approximately 50–70 percent across most upper deciles. Notably, the couple standard deviation nearly doubles in the top decile (97 percent versus 48 percent in the ninth), a jump not mirrored in the interdecile range (150 versus 111).

Second, saving rates are right-skewed in nearly all deciles for both household types. The positive skewness means that mean saving rates exceed median saving rates. This is visible in the kernel density estimates for the top decile (Figure 8), where the bulk of the distribution lies below zero for both singles and couples, but the couple distribution exhibits several distinct bumps in the right tail that pull the mean upward.

Figure 9 compares the overall saving rate distribution against a normal distribution with the same mean and standard deviation, separately for singles and couples. Both distributions exhibit a narrower central mass and heavier tails than the normal benchmark. The effect

**FIGURE 8.** Kernel Density of Saving Rates — Top Wealth Decile



*Notes: Kernel density estimates of saving rates in the top wealth decile using the Epanechnikov kernel. Gray denotes singles; black denotes couples. Dashed vertical lines indicate the mean saving rate for each group.*

is more pronounced for singles, whose distribution has a sharper peak and more extreme outliers. Couples display higher positive saving rates overall, yet their right tail is less extreme than that of singles. In both cases, the right skewness documented in Table 8 is clearly visible.

These distributional features—declining dispersion, persistent right skewness, and heavy tails—are consistent with findings from administrative data in other countries. Bach et al. (2017) document a similar pattern in Swedish administrative records, where saving rates at the top of the wealth distribution exhibit high dispersion and positive skewness. Fagereng et al. (2019) report comparable results using Norwegian tax data. The alignment between our HRS-based findings and these administrative studies, which do not suffer from the survey measurement issues inherent in the HRS, supports the robustness of the patterns documented here.

#### **B.4.5 The Role of Wealth Returns**

Table 9 reports the correlation between saving rates and three components of income—total income, earnings, and capital income—computed within each wealth decile.<sup>11</sup>

The key finding is the contrast between earnings and capital income. The correlation between saving rates and earnings is weak and flat: it remains below 0.2 in nearly all deciles for both household types and shows no systematic pattern across the wealth distribution. Earnings levels do not appear to drive variation in saving behavior within any wealth bracket.

Capital income tells a different story. For couples, the correlation with capital income rises from 0.08 in the bottom decile to 0.41 in the ninth decile, with the top three deciles all exceeding 0.28. For singles, a similar upward pattern is present, with the correlation reaching 0.30 in the eighth and ninth deciles. The total income correlation mirrors the capital income pattern rather than the earnings pattern, confirming that capital income is the dominant component driving the income-saving relationship at higher wealth levels.

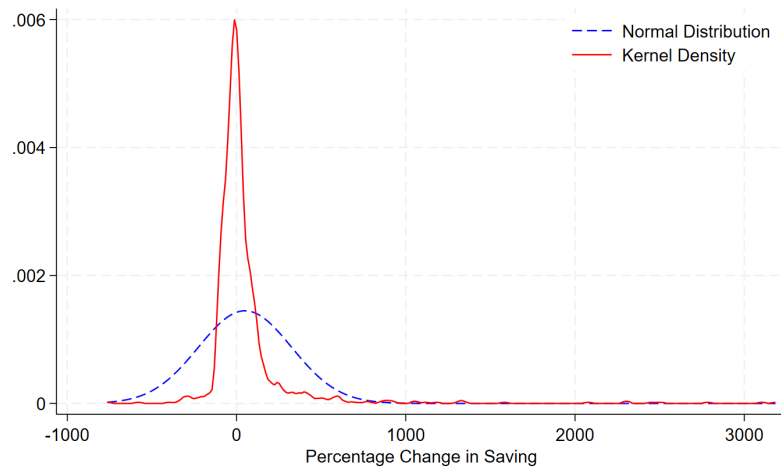
These results are consistent with the portfolio composition documented in Table 6: as house-

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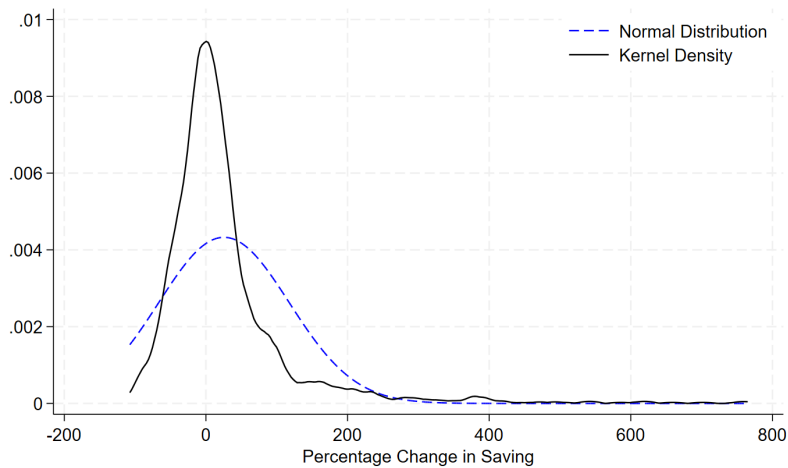
<sup>11</sup>All income variables enter in logs. For households with zero earnings, the log is replaced with the sample minimum of the log-transformed variable.

**FIGURE 9.** Kernel Density of Saving Rates vs. Normal Distribution

*a. Singles*



*b. Couples*



*Notes: Kernel density estimates of saving rates (solid) compared to a normal distribution with the same mean and standard deviation (dashed). Panel (a) shows singles; panel (b) shows couples. Epanechnikov kernel.*

**TABLE 9.** Correlation of Saving Rates with Income Components

|          | Total Income |         | Earnings |         | Capital Income |         |
|----------|--------------|---------|----------|---------|----------------|---------|
|          | Singles      | Couples | Singles  | Couples | Singles        | Couples |
| P0–P10   | −0.01        | 0.14    | −0.08    | 0.16    | 0.20           | 0.08    |
| P10–P20  | 0.11         | 0.16    | 0.13     | 0.09    | 0.23           | 0.35    |
| P20–P30  | 0.01         | 0.19    | 0.10     | 0.22    | 0.02           | 0.18    |
| P30–P40  | 0.17         | 0.29    | −0.03    | 0.13    | 0.28           | 0.34    |
| P40–P50  | 0.10         | 0.25    | 0.10     | 0.11    | 0.08           | 0.31    |
| P50–P60  | 0.30         | 0.08    | 0.16     | 0.00    | 0.34           | 0.19    |
| P60–P70  | 0.12         | 0.21    | 0.07     | 0.06    | 0.12           | 0.17    |
| P70–P80  | 0.27         | 0.23    | 0.15     | 0.09    | 0.30           | 0.20    |
| P80–P90  | 0.29         | 0.25    | −0.10    | 0.00    | 0.30           | 0.41    |
| P90–P100 | 0.24         | 0.34    | −0.01    | 0.10    | 0.25           | 0.28    |

*Notes: Pairwise correlations between the saving rate and the natural log of each income component, computed within each wealth decile separately for singles and couples.*

hold wealth increases, the portfolio shifts toward return-generating assets (financial holdings, real estate, business equity), and the saving rate becomes more closely tied to the returns on those assets. This suggests that wealth returns—rather than labor income—are the primary driver of saving heterogeneity at the top of the wealth distribution, a finding that motivates the theoretical mechanism developed in the main text.

## B.5 Summary

This appendix documented the distribution of wealth, saving, and saving rates among U.S. households aged 55–64, using the Health and Retirement Study. Wealth is highly concentrated: the top decile holds 63 percent of total wealth among singles and 52 percent among couples. The composition of wealth shifts systematically across the distribution, from housing and transportation at the bottom to financial assets, pensions, real estate, and business equity at the top. Couples hold more diversified portfolios than singles at every wealth level. Saving behavior differs sharply between household types. For couples, saving is positive throughout the distribution and increases monotonically with wealth, reaching over \$1 mil-

lion in the top decile. Saving rates decline from the bottom through the middle deciles but flatten and rise at the top. This pattern contradicts the prediction of standard income-fluctuation models, in which saving rates converge to zero or become negative as wealth increases. For singles, saving levels decline at the top of the distribution, and saving rates decrease monotonically from the third decile onward.

The saving rate distribution exhibits declining dispersion with wealth, persistent right skewness across nearly all deciles, and heavy tails relative to the normal distribution. These features are consistent with the patterns documented in Swedish and Norwegian administrative data by [Bach et al. \(2017\)](#) and [Fagereng et al. \(2019\)](#), suggesting that the distributional properties observed in the HRS are not driven by survey measurement issues.

Decomposing the correlation between saving rates and income reveals that earnings explain little of the variation in saving across wealth brackets. Capital income, by contrast, is more strongly correlated with saving rates, and this correlation rises in higher deciles—particularly for couples. This pattern aligns with the return heterogeneity documented in the administrative data literature and with the compositional shift toward return-generating assets at the top of the wealth distribution.

## C Imputing Total Health Expenditure

This appendix develops the imputation procedure used to construct total medical expenditure in the HRS and establishes its theoretical properties. Building on [?](#), out-of-pocket spending—the only medical cost measure available in the HRS—captures only the household’s direct cost share and compresses the true spending gradient across the wealth distribution. Using the Medical Expenditure Panel Survey (MEPS), in which total medical expenditure is verified directly by healthcare providers, we estimate the relationship between out-of-pocket and total spending and use it to impute total medical expenditure in the HRS. This appendix also proves that using the imputed variable as the dependent variable in an OLS regression preserves consistency of the slope coefficients under standard exogeneity assumptions. Finally, it documents that total medical spending rises monotonically with wealth for couples, with the out-of-pocket gradient roughly half as steep—a gap exceeding 40 percentage points at the top of the wealth distribution.

## C.1 Imputation Procedure

Consider the relationship between out-of-pocket spending and total medical spending:

$$p_i = D_i\beta + \gamma h_i + \epsilon_i, \quad i \in \{M, H\}. \quad (30)$$

Here  $p$  denotes out-of-pocket spending,  $D$  is a vector of demographic and other controls,  $h$  is total medical spending, and  $\epsilon$  is an error term. The index  $i \in \{M, H\}$  denotes the donor (MEPS) and main (HRS) samples, respectively. Rearranging (30) yields the imputed total spending:

$$\hat{h}_i = \frac{p_i - D_i\hat{\beta}}{\hat{\gamma}}, \quad (31)$$

where  $\hat{\beta}$  and  $\hat{\gamma}$  are estimated in MEPS and  $\hat{\gamma} \neq 0$ .

Substituting (30) into (31) yields the relationship between the true and imputed values:

$$\hat{h}_i = D_i \frac{(\beta - \hat{\beta})}{\hat{\gamma}} + \frac{\gamma}{\hat{\gamma}} h_i + \frac{\epsilon_i}{\hat{\gamma}} \quad (32)$$

The following assumptions ensure consistency of  $\hat{\beta}$  and  $\hat{\gamma}$ :

**ASSUMPTION C.1** *The donor (MEPS) and main (HRS) samples are drawn from the same underlying population with respect to medical spending. Moreover,*

$$[\epsilon] = 0, \quad [D'\epsilon] = 0, \quad [h\epsilon] = 0.$$

Under Assumption C.1, the imputed values are mean-consistent within the donor sample:

$$\begin{aligned} \bar{\hat{h}}_M &= \mu_h, \\ s_{\hat{h}_M}^2 &= \sigma_h^2 + \frac{1}{\gamma^2} \sigma_{\epsilon_M}^2. \end{aligned} \quad (33)$$

The sample variance exhibits a persistent positive bias due to imputation noise, even though the mean is unbiased.

When applied to the HRS, the imputed mean may diverge from the MEPS benchmark due

to differences in out-of-pocket spending distributions:

$$\bar{h}_H = \mu_h + \frac{1}{\gamma} (\bar{p}_H - \bar{p}_M) - \frac{1}{\gamma} (\bar{D}_H - \bar{D}_M) \beta \quad (34)$$

The following result provides a simple adjustment:

**THEOREM C.1** *Define  $\hat{h}_a = \hat{h}_H - \frac{1}{\gamma}(\bar{p}_H - \bar{p}_M)$ . If the underlying demographic distributions in MEPS and HRS coincide and Assumption C.1 holds, then  $\hat{h}_a$  is mean consistent:  $\bar{\hat{h}}_a = \mu_h$ .*

## C.2 Incorporating Imputed Variables into OLS Estimation

A key concern is whether using the imputed variable as the dependent variable in a subsequent regression preserves consistency. Consider the system:

$$\begin{aligned} p &= h \gamma + D \beta_1 + \epsilon \\ h &= w \alpha + D \beta_2 + u \end{aligned} \quad (35)$$

where  $w$  denotes wealth. Define the residual maker  $M_D = I - D(D'D)^{-1}D'$  and the residualized variables  $\ddot{x} = M_D x$ . The OLS estimator of  $\alpha$  using imputed  $\hat{h}$  is consistent provided that residualized wealth is orthogonal to both residualized error terms.

Since out-of-pocket spending is a share of total health spending, wealth affects out-of-pocket spending exclusively through total health expenditure. Conditional on the controls  $D$ , there is no direct channel from wealth to the structural error in the imputation equation, which justifies the exclusion restriction.

**ASSUMPTION C.2** *Both MEPS and HRS data sets are from the same underlying population, and (35) governs the relationships between total health spending, out-of-pocket spending, and wealth. Further, in addition to Assumption C.1:*

1.  $E[h'\epsilon] = E[w'u] = 0$
2.  $E[D'\epsilon] = E[D'u] = 0$

**THEOREM C.2** *Under Assumption C.2, the OLS estimator  $\hat{\alpha}$  from regressing imputed  $\hat{h}$  on wealth  $w$  and controls  $D$  is consistent for  $\alpha$ .*

*Proof.* The proof proceeds in three steps. First, under Assumption C.2,  $\hat{\gamma}$  and  $\hat{\beta}_1$  are consistent by the law of large numbers and the orthogonality conditions. Second, the imputed value can be decomposed as  $\ddot{h}_i = \ddot{h}_i(\gamma/\hat{\gamma}) + \ddot{\epsilon}_i/\hat{\gamma}$ . Third, substituting into the OLS formula and using the orthogonality conditions  $\mathbb{E}[w'u] = 0$ ,  $\mathbb{E}[D'u] = 0$ ,  $\mathbb{E}[D'\epsilon] = 0$ , and  $\mathbb{E}[w'\epsilon] = 0$  (the last following from  $h = w\alpha + D\beta_2 + u$  combined with  $\mathbb{E}[h'\epsilon] = 0$ ), the bias terms vanish, yielding  $\hat{\alpha} = \alpha$ .  $\square$

### C.3 Data and Imputation Validation

The Medical Expenditure Panel Survey (MEPS) serves as the donor dataset. MEPS collects data on specific health services, their costs, and how they are paid for, with reports verified directly by healthcare providers. This provider-verified structure alleviates concerns about measurement error bias and motivates the use of MEPS as the donor dataset. The HRS is the main dataset in which we apply the imputation.

The specification omits an intercept term, since zero total spending mechanically implies zero out-of-pocket spending. The log transformation linearizes the concave relationship between out-of-pocket and total spending, and the demographic controls subsume the role of the intercept.

We restrict the sample to individuals aged 51–84, with out-of-pocket spending at or below \$20,000, positive total and out-of-pocket spending, no nursing home residence exceeding one night, and racial classification as Black or White. The analysis covers 2006–2016, corresponding to waves for which both datasets are jointly available.

Table 10 compares demographic means across the two datasets. Gender and racial compositions are closely aligned. Differences are more pronounced for marital status (HRS respondents are 6–9 percentage points more likely to be married) and education (MEPS respondents are 5–8 percentage points more likely to hold a university degree), but their effect on the imputation is negligible given that the corresponding regression coefficients in the imputation equation are small.

Table 11 compares true and imputed total expenditures within MEPS. The imputation successfully reproduces mean values, with differences of less than 0.1 log points in all years. Standard deviations exhibit a persistent positive bias consistent with (33), with the gap

**TABLE 10.** Comparing Demographic Means

| HRS               |       |       |       |       |       |       | MEPS              |       |       |       |       |       |       |
|-------------------|-------|-------|-------|-------|-------|-------|-------------------|-------|-------|-------|-------|-------|-------|
|                   | 2006  | 2008  | 2010  | 2012  | 2014  | 2016  |                   | 2006  | 2008  | 2010  | 2012  | 2014  | 2016  |
| Sex               | 0.45  | 0.46  | 0.46  | 0.46  | 0.46  | 0.46  | Sex               | 0.46  | 0.47  | 0.47  | 0.47  | 0.47  | 0.47  |
| Race              | 0.92  | 0.91  | 0.91  | 0.91  | 0.92  | 0.91  | Race              | 0.91  | 0.90  | 0.90  | 0.89  | 0.89  | 0.89  |
| Married           | 0.72  | 0.71  | 0.71  | 0.70  | 0.70  | 0.71  | Married           | 0.66  | 0.64  | 0.64  | 0.63  | 0.61  | 0.64  |
| University Degree | 0.28  | 0.29  | 0.32  | 0.33  | 0.34  | 0.35  | University Degree | 0.34  | 0.36  | 0.39  | 0.40  | 0.42  | 0.41  |
| Highschool Degree | 0.87  | 0.88  | 0.90  | 0.91  | 0.92  | 0.93  | Highschool Degree | 0.86  | 0.86  | 0.88  | 0.88  | 0.85  | 0.89  |
| Age               | 63.84 | 65.06 | 62.82 | 63.92 | 65.24 | 63.88 | Age               | 63.46 | 63.49 | 63.32 | 63.43 | 63.62 | 64.20 |

*This table compares main regressors in the imputation process. Sex takes the value 1 if the respondent is male, and zero otherwise. Race takes the value of 1 if the respondent is white. Married represents the share of married couples and the last two variables illustrate the share of respondents with a university and high school degree.*

widening over time from approximately 0.3 in 2006 to 0.7 in 2016—likely reflecting compositional shifts from HRS refreshment cohorts.

**TABLE 11.** True vs Imputed Total Expenditure — MEPS

|      | Data |     |     |          | Imputation |     |     |          |
|------|------|-----|-----|----------|------------|-----|-----|----------|
|      | mean | sd  | iqr | skewness | mean       | sd  | iqr | skewness |
| 2006 | 8.6  | 1.4 | 1.6 | -0.10    | 8.6        | 1.7 | 2.0 | -0.19    |
| 2008 | 8.7  | 1.4 | 1.7 | -0.13    | 8.8        | 1.8 | 2.1 | -0.20    |
| 2010 | 8.7  | 1.5 | 1.9 | -0.11    | 8.7        | 2.0 | 2.4 | -0.16    |
| 2012 | 8.6  | 1.6 | 1.9 | -0.12    | 8.7        | 2.0 | 2.5 | -0.19    |
| 2014 | 8.7  | 1.5 | 1.9 | -0.13    | 8.8        | 2.1 | 2.5 | -0.22    |
| 2016 | 8.8  | 1.6 | 1.9 | -0.11    | 8.9        | 2.3 | 2.8 | -0.17    |

*This table summarizes distribution moments of actual and imputed total medical expenditure in MEPS. All values are in logs. Skewness is quantile-based.*

When applied to the HRS, the unadjusted imputed means diverge from the MEPS benchmark, but the adjustment from Theorem C.1 successfully removes this divergence, confirming that the imputation targets the correct population moments after accounting for cross-survey differences in out-of-pocket spending.

## C.4 Medical Spending–Wealth Gradient

To characterize the medical spending–wealth gradient, we estimate the following specification separately for couple and single households:

$$\hat{h}_{it} = \alpha + \sum_{d=2}^{10} \theta_d \cdot \mathbf{1}[W_{it} \in \text{decile } d] + D_{it}\beta + \delta_t + u_{it} \quad (36)$$

where  $\hat{h}_{it}$  denotes imputed log total medical expenditure,  $W_{it}$  is household net worth,  $D_{it}$  collects demographic controls, and  $\delta_t$  are wave fixed effects. Standard errors are clustered at the household level.

Table 12 reports the results. For couples, the total spending gradient is monotonically positive, rising from 27 percent at the second decile to 89 percent at the tenth decile relative to the poorest households. The out-of-pocket gradient is substantially flatter, reaching only 47 percent at the top decile. The gap between the two measures exceeds 40 percentage points at the top of the wealth distribution.

For singles, both measures exhibit a U-shaped pattern at the bottom of the wealth distribution. The two gradients track each other more closely than in the couple sample: at the tenth decile, total spending is 42 percent above the reference group compared to 35 percent for out-of-pocket, a gap of only 7 percentage points. The narrower gap for singles is consistent with less generous insurance coverage among single-person households.

These findings indicate that studies relying solely on out-of-pocket expenditure may substantially understate the relationship between medical spending and wealth, particularly for couples. The imputed total medical expenditure constructed here provides the measure used to estimate the health production function in the main text (Table 1).

**TABLE 12.** Medical Spending by Wealth Decile: Total vs. Out-of-Pocket

| Decile | Couples                   |              |            | Singles                   |              |            |
|--------|---------------------------|--------------|------------|---------------------------|--------------|------------|
|        | Median Wealth<br>(\$000s) | Total<br>(%) | OOP<br>(%) | Median Wealth<br>(\$000s) | Total<br>(%) | OOP<br>(%) |
| 1      | 0.4                       | ref.         | ref.       | −7.5                      | ref.         | ref.       |
| 2      | 41.5                      | 26.7%        | 15.1%      | 0.0                       | −46.3%       | −39.6%     |
| 3      | 91.2                      | 28.2%        | 19.2%      | 3.7                       | −24.6%       | −16.8%     |
| 4      | 151.0                     | 33.6%        | 16.4%      | 17.1                      | 0.6%         | 0.4%       |
| 5      | 225.1                     | 28.4%        | 17.9%      | 46.0                      | 8.2%         | 5.5%       |
| 6      | 325.2                     | 49.5%        | 23.6%      | 84.1                      | 29.4%        | 21.6%      |
| 7      | 470.8                     | 49.4%        | 20.2%      | 139.8                     | 15.0%        | 14.6%      |
| 8      | 674.2                     | 46.8%        | 21.0%      | 233.4                     | 9.7%         | 12.0%      |
| 9      | 1,062.8                   | 64.1%        | 27.3%      | 424.2                     | 31.4%        | 27.0%      |
| 10     | 2,251.1                   | 88.9%        | 47.3%      | 977.7                     | 42.1%        | 34.8%      |

Notes: Total and OOP columns report  $100 \times (e^{\hat{\theta}_d} - 1)$ , the percentage difference in spending relative to decile 1. Total refers to imputed total medical expenditure; OOP refers to observed out-of-pocket medical expenditure. Both regressions include wave fixed effects and demographic controls. Standard errors clustered at the household level.